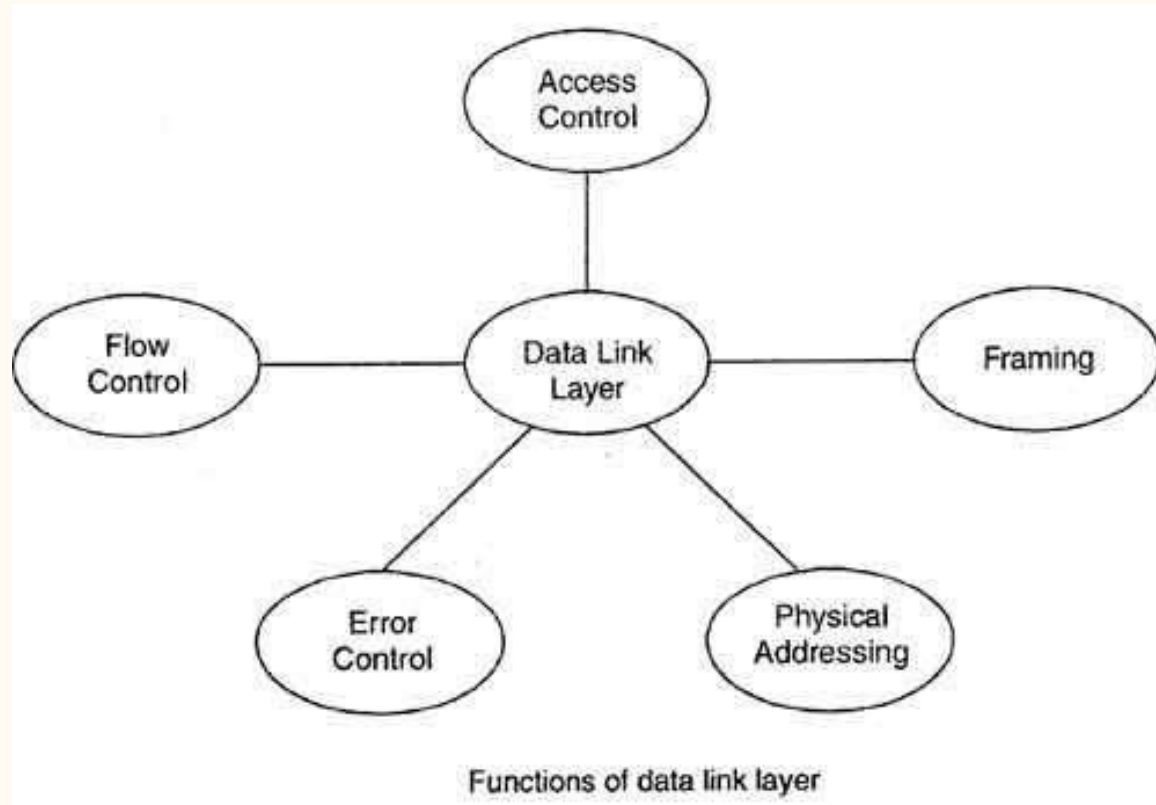


Data Link Layer

Er. Sudarshan Devkota

Functions of Data Link Layer

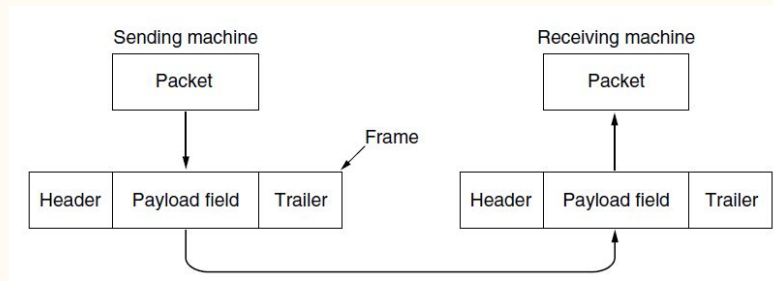


Functions of Data Link Layer

1.Framing: Data-link layer takes packets from Network Layer and encapsulates them into Frames. Then, it sends each frame bit-by-bit on the hardware. At receiver' end, data link layer picks up signals from hardware and assembles them into frames.

2.Addressing: Data-link layer provides layer-2 hardware addressing mechanism. Hardware address is assumed to be unique on the link. It is encoded into hardware at the time of manufacturing.

3.Synchronizatiion: When data frames are sent on the link, both machines must be synchronized in order to transfer to take place.



Functions of Data Link Layer

4.Error Control: Sometimes signals may have encountered problem in transition and the bits are flipped. These errors are detected and attempted to recover actual data bits. It also provides error reporting mechanism to the sender.

5.Flow Control: Stations on same link may have different speed or capacity. Data-link layer ensures flow control that enables both machines to exchange data on same speed.

6.Multi-Access: When host on the shared link tries to transfer the data, it has a high probability of collision. Data-link layer provides mechanism such as CSMA/CD to equip capability of accessing a shared media among multiple Systems.

7.Reliable delivery When a link-layer protocol provides reliable delivery service, it guarantees to move each network-layer datagram across the link without error.

Framing

Framing is the process of dividing a continuous stream of bits received from the network layer into manageable units called frames.

Parts of a Frame:

A frame has the following parts:

- **Frame Header:** It contains the source and the destination addresses of the frame.
- **Payload field:** It contains the message to be delivered.
- **Trailer:** It contains the error detection and error correction bits.
- **Flag:** It marks the beginning and end of the frame.



Methods of Framing: Character count

In this method, the sender includes a count field at the beginning of each frame. This field tells the receiver how many bytes are present in the frame.

How it works

- Sender calculates the length of the data.
- This length is placed at the start of the frame.
- Receiver reads the length field.
- It then extracts exactly that many bytes.

Methods of Framing: Character count

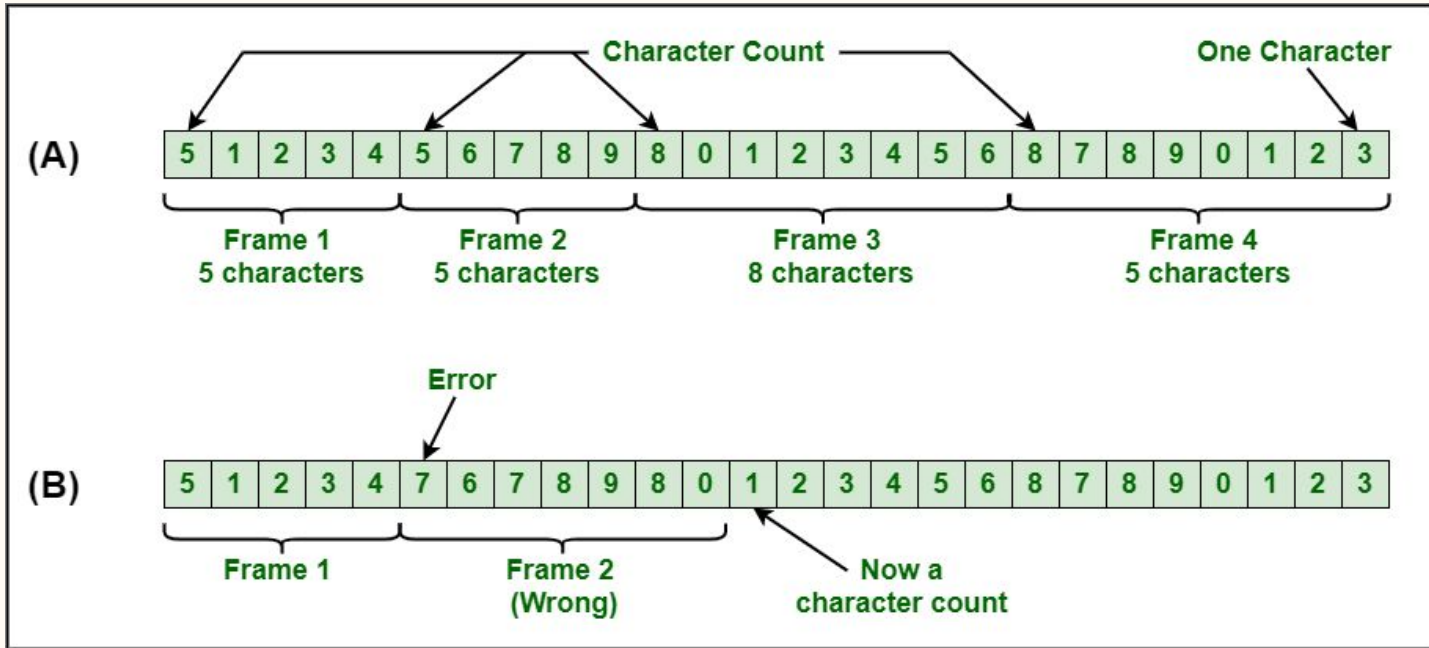
Advantages:

- Very easy to implement
- No extra characters or bits are needed for marking boundaries
- Efficient when transmission errors are rare

Disadvantages:

- If the length field is corrupted, the receiver loses track of frame boundaries
- One error can affect not just one frame, but all subsequent data
- Not suitable for noisy communication channels

Methods of Framing: Character count



A Character Stream

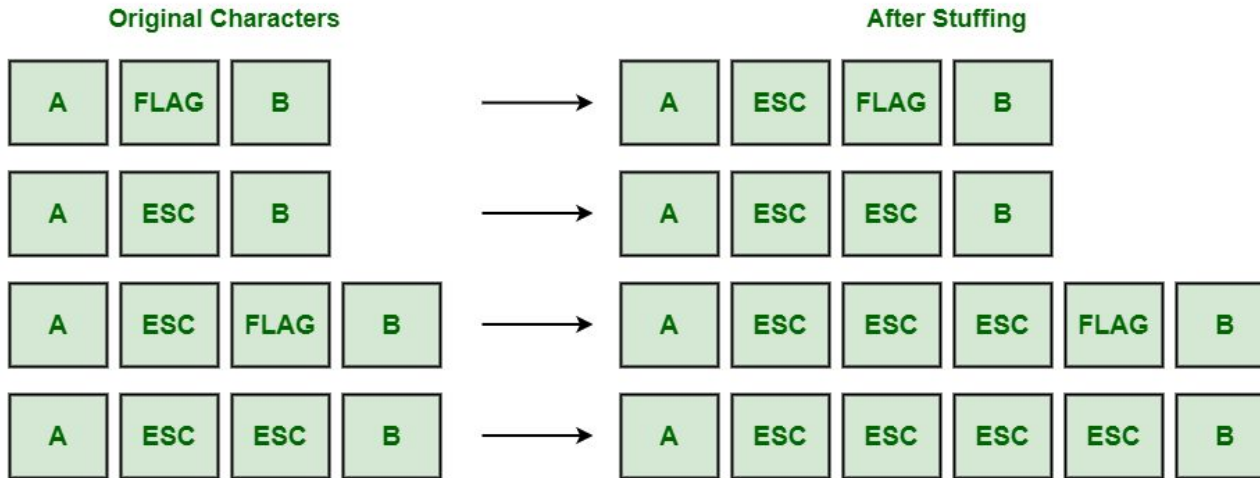
(A) Without Errors

(B) With one Error

Methods of Framing: Byte Stuffing

- In this method, start and end of frame are recognized with the help of flag bytes.
- Each frame starts with and ends with a flag byte.
- Two consecutive flag bytes indicate the end of one frame and start of the next one.
- A serious problem occurs with this method when flag byte's bit pattern occurs in the data.
- This is solved by inserting special escape byte(ESC) just before each flag byte in the data.

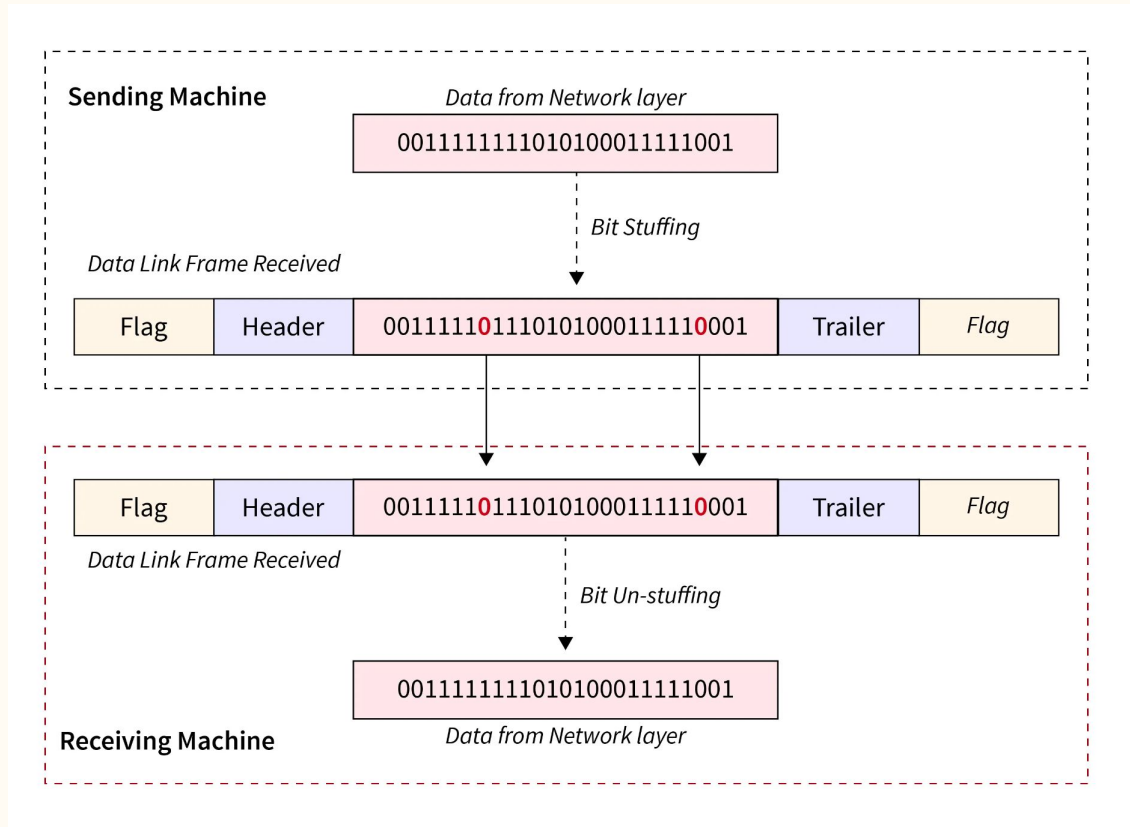
Methods of Framing: Byte Stuffing



Methods of Framing: Bit Stuffing

- Frames are separated using a special flag bit pattern: **01111110** (marks start and end).
- The method works at the bit level, so data can be of any length and is not tied to characters.
- To prevent the flag pattern from appearing inside data, a 0 is inserted after every five consecutive 1s.
- At the receiver, this extra 0 is removed, restoring the original data.
- This ensures correct frame boundary detection without confusion between data and control bits.

Methods of Framing: Bit Stuffing

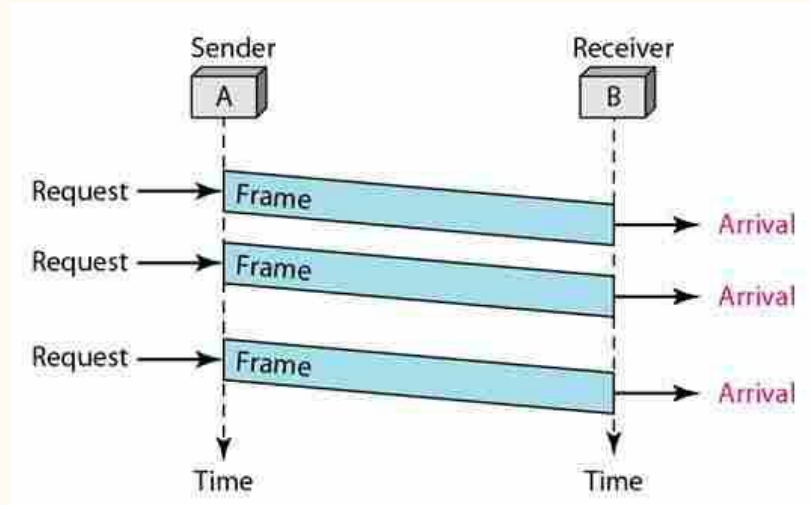


Flow Control

- We must deal with the issue where the sender is sending data at a higher rate than the receiver can receive the data.
- Flow Control is a technique for speed-matching of transmitter and receiver.
- Flow control ensures that a transmitting station does not overflow a receiving station with data.

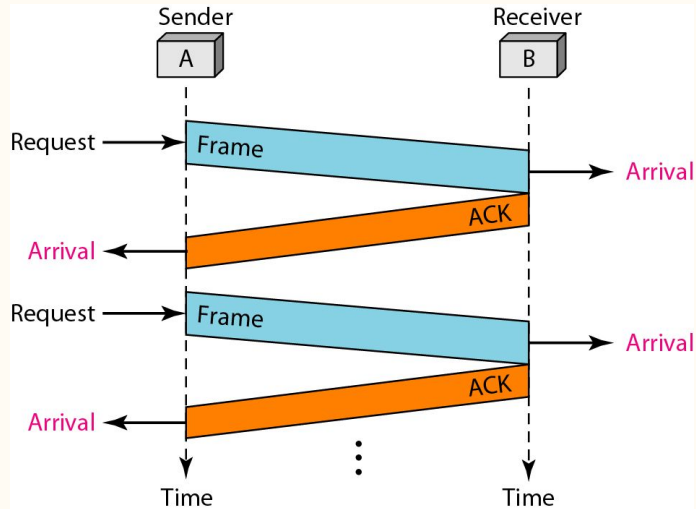
Flow Control: Simplest

- Simplest Protocol is one that has no flow or error control and it is a unidirectional protocol in which data frames are traveling in only one direction—from the sender to receiver.
- We assume that the receiver can immediately handle any frame it receives with a processing time that is small enough to be negligible.



Flow Control: Stop And Wait

- The sender transmits exactly one frame.
- The sender then stops and waits, it does not send anything else.
- The receiver processes the frame and sends back an ACK.
- Only after the sender receives the ACK does it send the next frame.



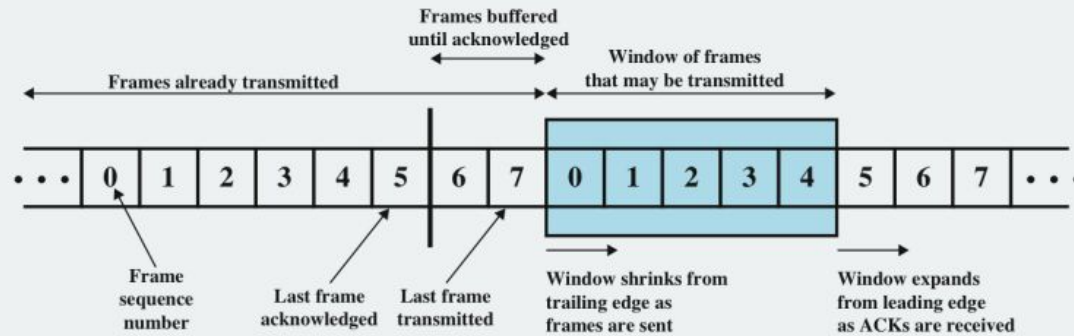
Flow Control: Sliding Window

This protocol improves the efficiency of stop and wait protocol by allowing multiple frames to be transmitted before receiving an acknowledgment.

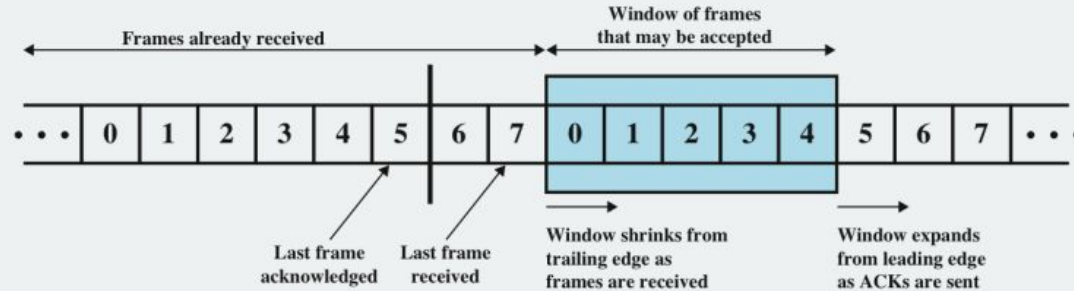
Working principle:

- Both the sender and the receiver have finite sized buffers called windows.
- The sender and the receiver agree upon the number of frames to be sent based upon the buffer size.
- The sender sends multiple frames in a sequence, without waiting for acknowledgment.
- When its sending window is filled, it waits for acknowledgment.
- On receiving acknowledgment, it advances the window and transmits the next frames, according to the number of acknowledgments received.

Flow Control: Sliding Window



(a) Sender's perspective



(b) Receiver's perspective

Figure 7.3 Sliding-Window Depiction

Error Control

- Error control makes sure that all frames are eventually delivered to the network layer at the destination in proper order.
- It is based upon the principle of request for automatic retransmission (ARQ) of the missing, lost or damaged frame.

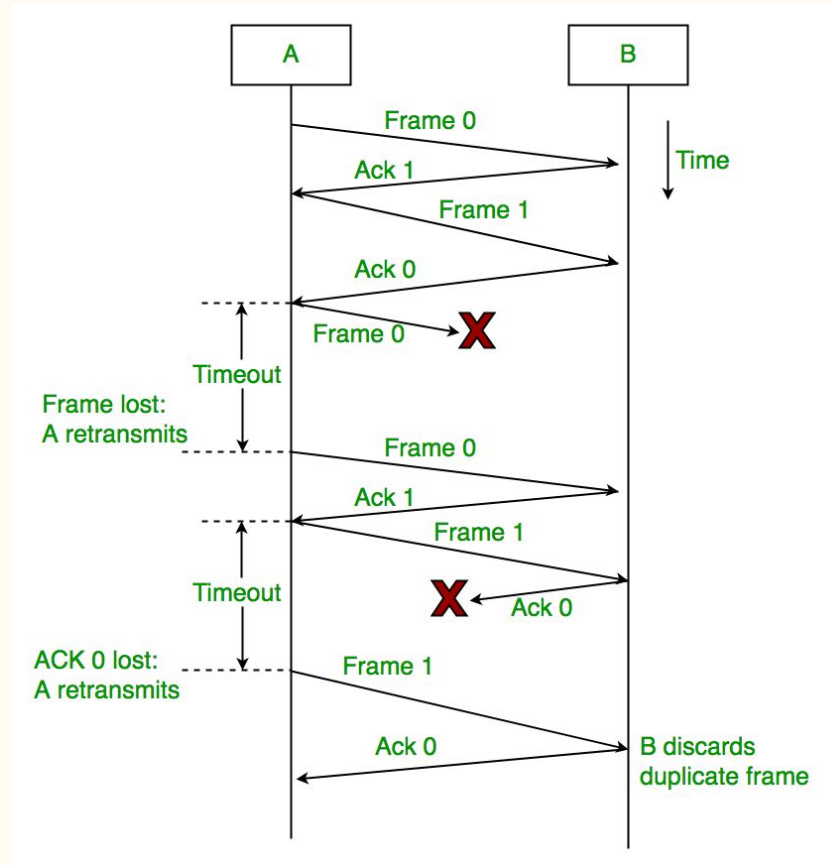
ARQ system has three types:

- 1) Stop and Wait ARQ
- 2) Go Back N ARQ
- 3) Selective Repeat

Error Control : Stop and Wait ARQ

- The sender sends one frame, stops until it receives confirmation from the receiver and then sends the next frame.
- For retransmission, it keeps a copy of the frame that was sent in case of lost or corrupted frame.
- The identification of data frame and ACK frame is 0 and 1 respectively.
- The sending device is equipped with a timer.
- If the sender does not receive ACK before the timeout, it retransmits the frame.

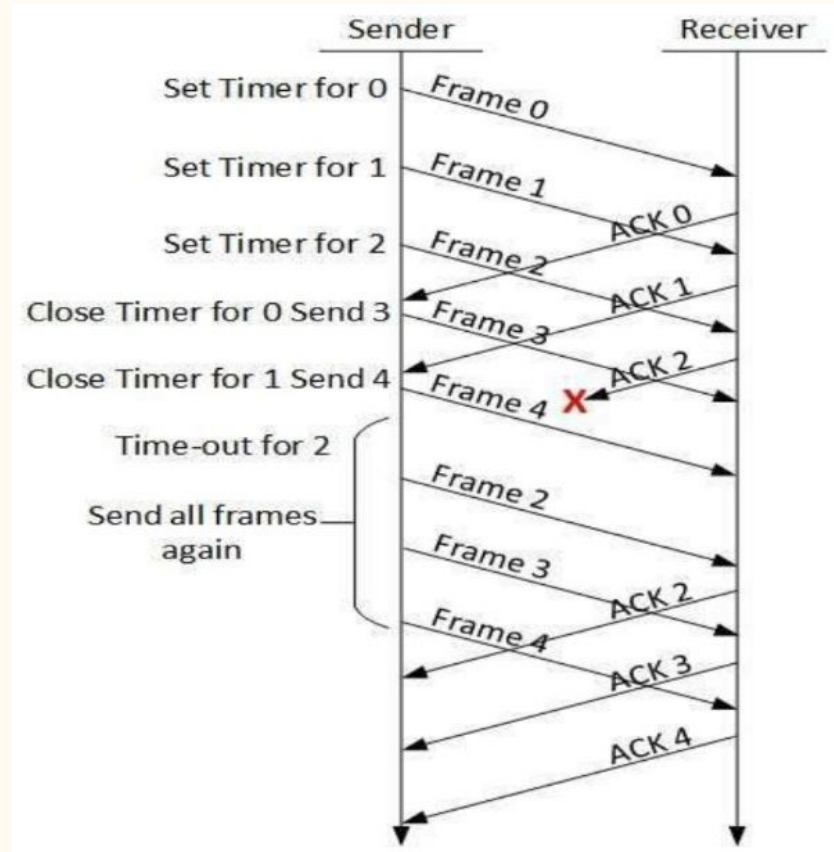
Error Control : Stop and Wait ARQ



Error Control : Go Back N ARQ

- It is based on sliding window flow control.
- The sending-window size enables the sender to send multiple frames without receiving the acknowledgement of the previous ones.
- The receiving-window enables the receiver to receive multiple frames and acknowledge them.
- The sender pushes out a full window of frames and waits for feedback.
- If everything is acknowledged (ACK), it moves on to the next set.
- However, if a frame is missing or comes back with a negative acknowledgement (NACK), the sender doesn't just fix that one mistake, it goes back and retransmits that frame and every single one that followed it.

Error Control : Go Back N ARQ

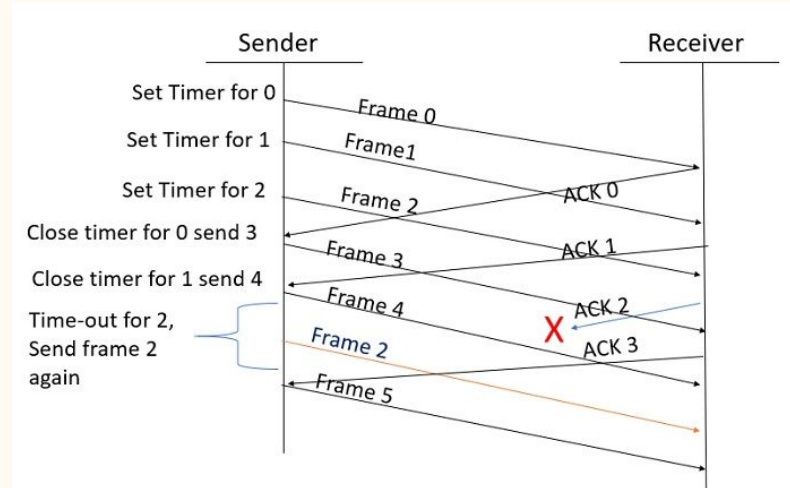


Error Control : Selective Repeat ARQ

As in Go Back N ARQ, a station may send a series of frames sequentially numbered with a maximum value.

But in Selective-Repeat ARQ, the receiver while keeping track of sequence numbers, buffers the frames in memory and sends NACK for only frame which is missing or damaged.

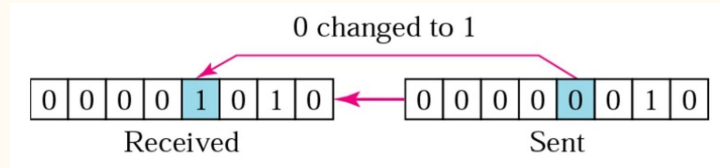
The sender in this case, sends only packet for which NACK is received.



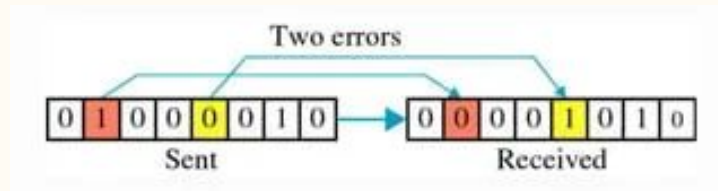
Error Control

Types of Error:

- 1) *Single bit error*: In a frame, there is only one bit, anywhere though, which is corrupt.

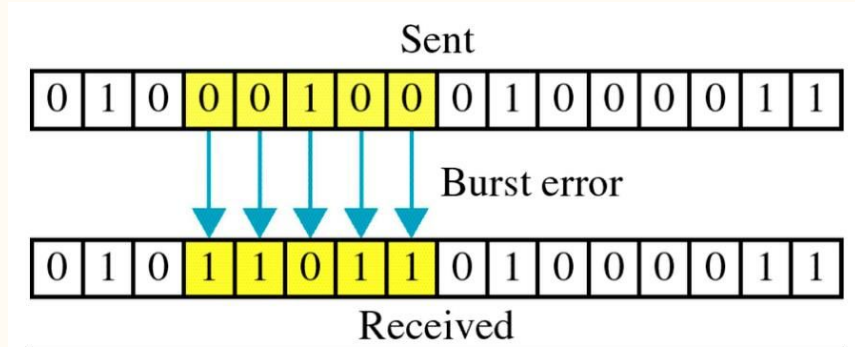


- 2) *Multiple bit error*: Frame is received with more than one bits in corrupted state.



Error Control

3) *Burst Error*: Frame contains more than 1 consecutive bits corrupted.



Error control can be done in two ways:

1. Error detection
2. Error correction

Error Detection: Parity Check

A parity bit is a simple error detection technique in which one extra bit is added to a data unit to make the number of 1s either even or odd.

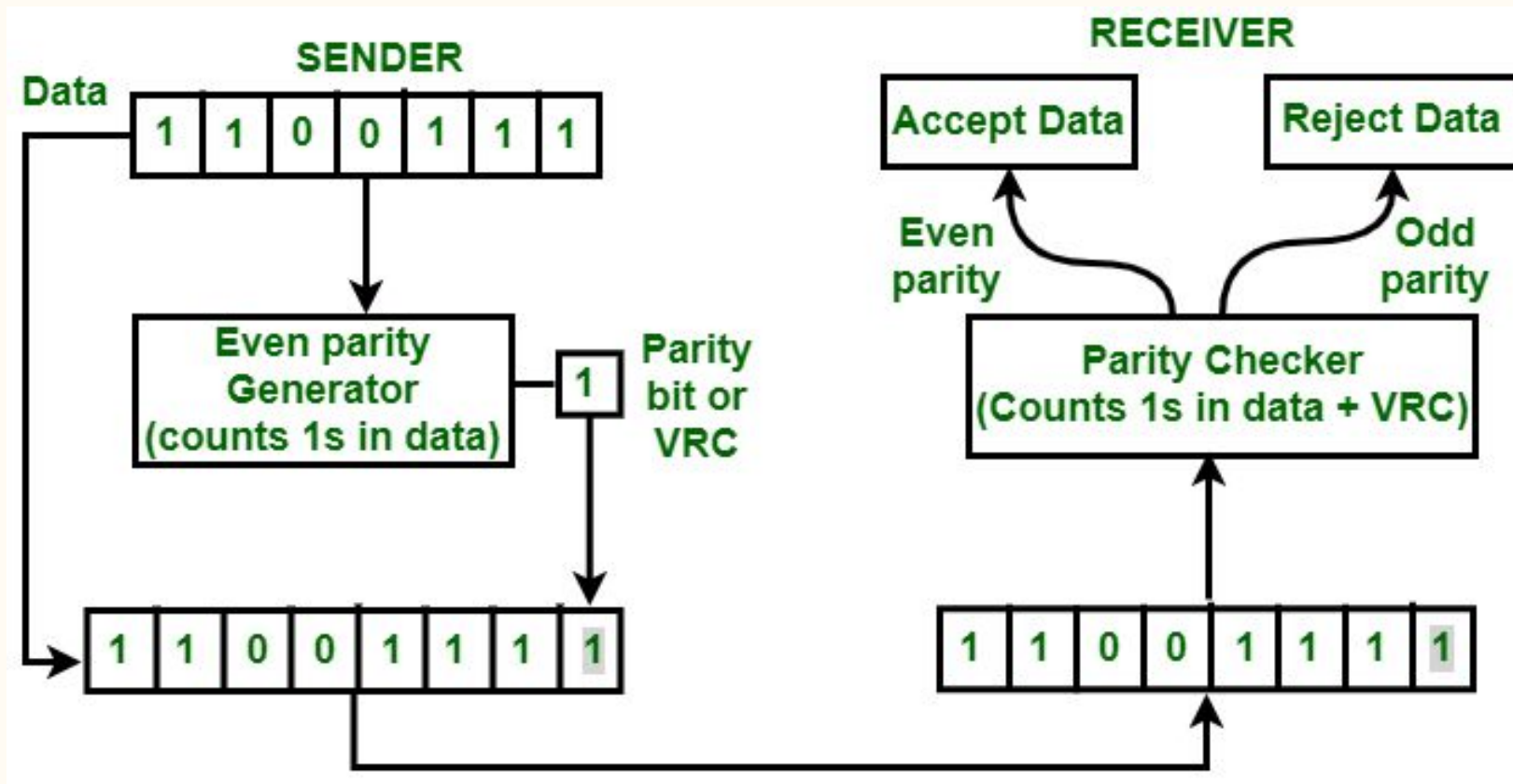
Working

- The sender counts the number of 1s in the data
- Based on the type of parity (even or odd), it adds a parity bit
- The receiver performs the same count and checks if the condition still holds

Types of Parity

1. *Even Parity*: Total number of 1s (including parity bit) must be even.
2. *Odd Parity*: Total number of 1s (including parity bit) must be odd

Error Detection: Parity Check



Error Detection: Checksum

At the Sender:

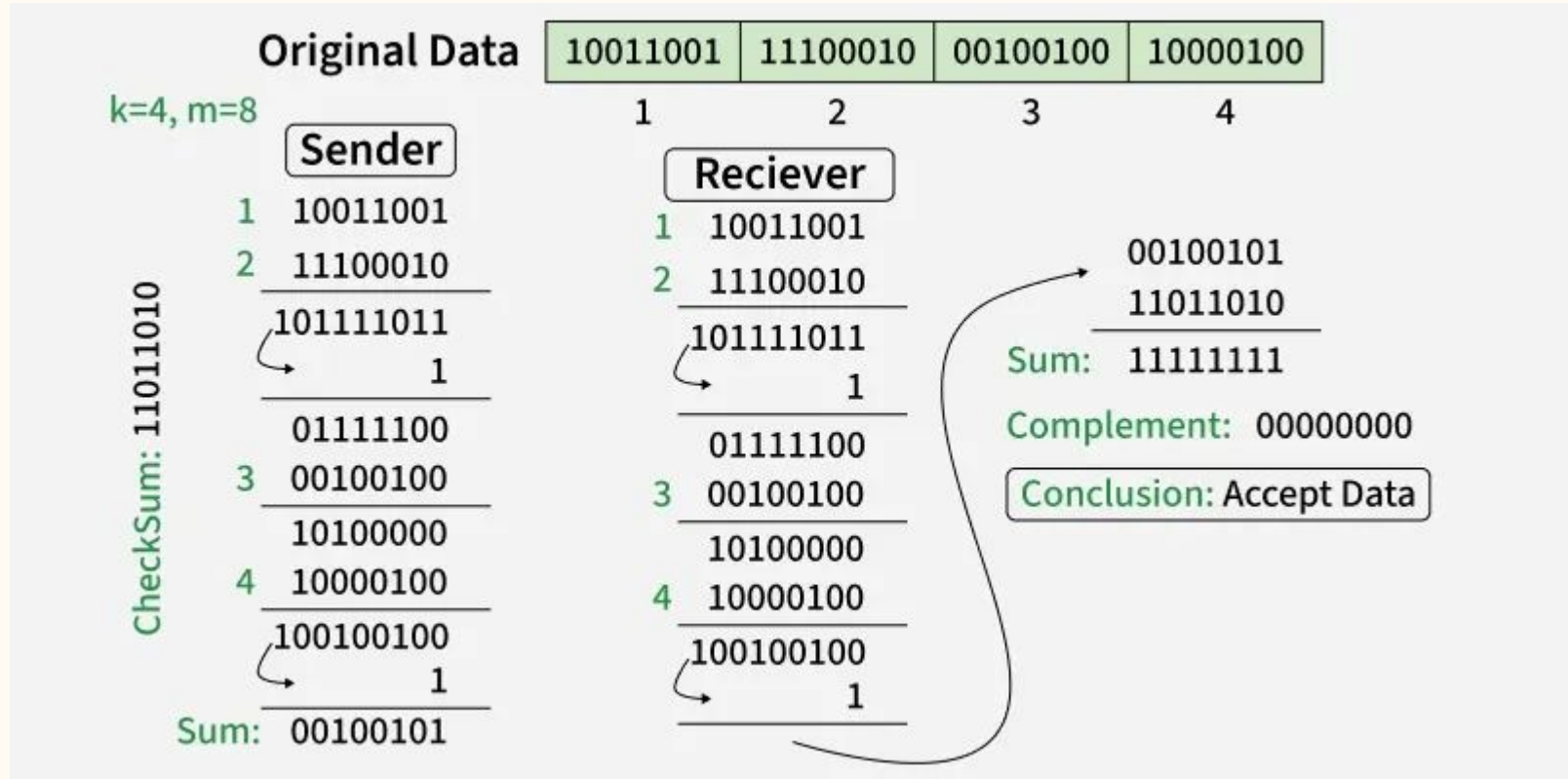
- The data is divided into equal-sized blocks of bits (such as 8-bit or 16-bit words).
- All the blocks are added together using binary addition.
- If a carry is generated beyond the most significant bit, it is wrapped around and added back to the result (end-around carry).
- This carry adjustment is repeated until no extra carry remains.
- The final sum is then complemented by inverting all bits to obtain the checksum.
- The checksum is appended to the original data and transmitted.

Error Detection: Checksum

At the Receiver:

- The received data, including the checksum, is divided into the same size blocks.
- All the blocks are added together using the same binary addition method.
- Any carry generated is wrapped around and added back to the result.
- The final sum is complemented by inverting all bits.
- If the result is all zeros, the data is accepted as correct; otherwise, an error is detected.

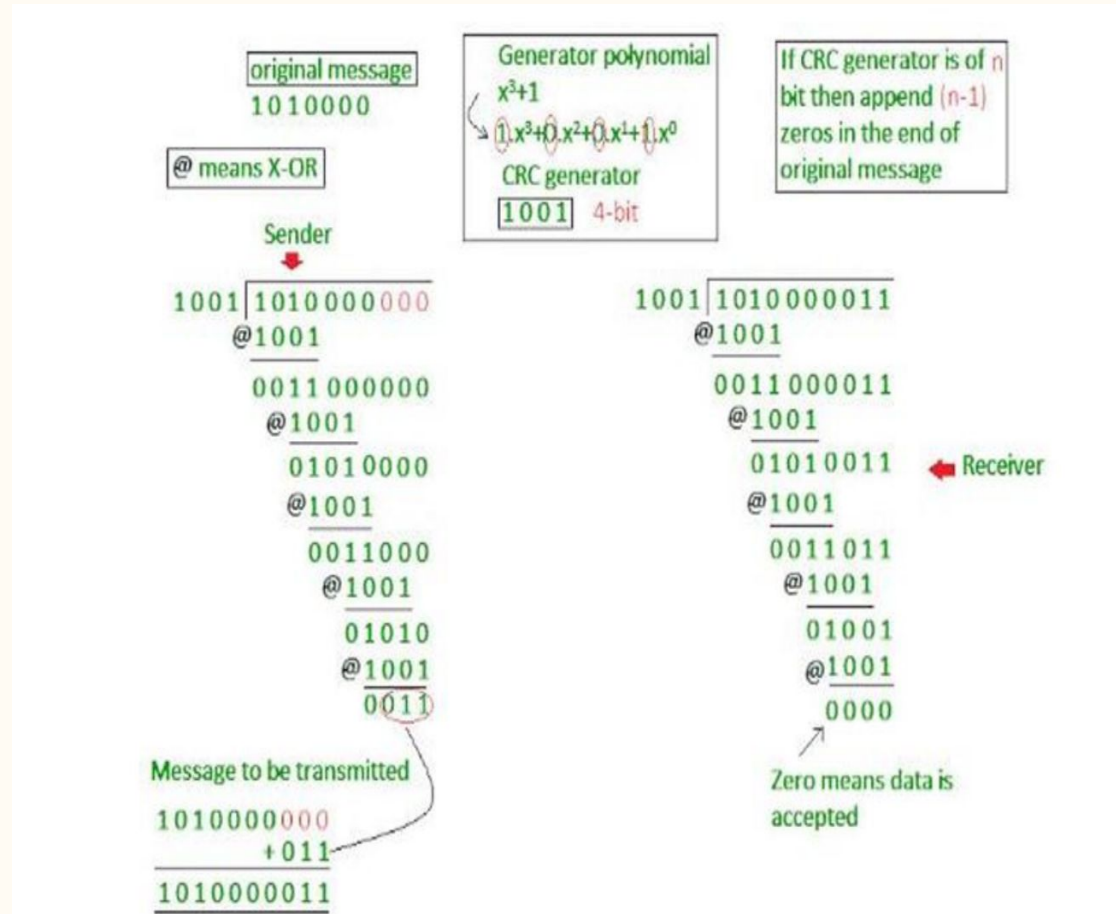
Error Detection: Checksum



Error Detection:Cyclic Redundancy Check(CRC)

- The sender and receiver agree on a generator polynomial (also called the divisor).
- The sender appends a remainder called the Frame Check Sequence (FCS)to the data before transmitting.
- The receiver divides the received frame by the same generator; if the remainder is zero, no error was detected.

Error Detection: Cyclic Redundancy Check (CRC)



Error Correction

- Error correction is the process of detecting and fixing errors that occur during data transmission over a communication channel.
- Error correction techniques add redundancy (extra bits) to the original data so that errors can be identified and corrected at the receiver.

Ways

1. Backward Error Correction (Retransmission)
2. Forward Error Correction
 - The sender adds redundant bits to the data before transmission.
 - The receiver uses these extra bits to detect and correct errors without asking for retransmission.

Error Correction: Hamming Code

Step 1: Write the bit positions starting from 1 in binary form (1, 10, 11, 100, etc).

Step 2: All the bit positions that are a power of 2 are marked as parity bits (1, 2, 4, 8, etc).

Step 3: All the other bit positions are marked as data bits.

Step 4: Each data bit is included in a unique set of parity bits, as determined its bit position in binary form:

- Parity bit 1 covers all the bits positions whose binary representation includes a 1 in the least significant position (1, 3, 5, 7, 9, 11, etc).
- Parity bit 2 covers all the bits positions whose binary representation includes a 1 in the second position from the least significant bit (2, 3, 6, 7, 10, 11, etc).

Error Correction: Hamming Code

- Parity bit 4 covers all the bits positions whose binary representation includes a 1 in the third position from the least significant bit (4-7, 12-15, 20-23, etc).
- Parity bit 8 covers all the bits positions whose binary representation includes a 1 in the fourth position from the least significant bit bits (8-15, 24-31, 40-47, etc).

Step 5: Since we check for even parity set a parity bit to 1 if the total number of ones in the positions it checks is odd. Set a parity bit to 0 if the total number of ones in the positions it checks is even.

Error Correction: Hamming Code

Position	R8	R4	R2	R1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1

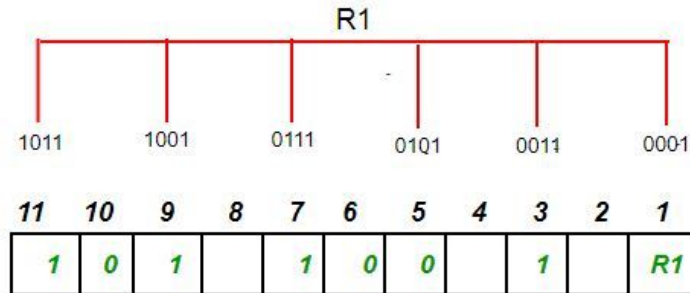
R1 -> 1,3,5,7,9,11
R2 -> 2,3,6,7,10,11
R3 -> 4,5,6,7
R4 -> 8,9,10,11

Error Correction: Hamming Code

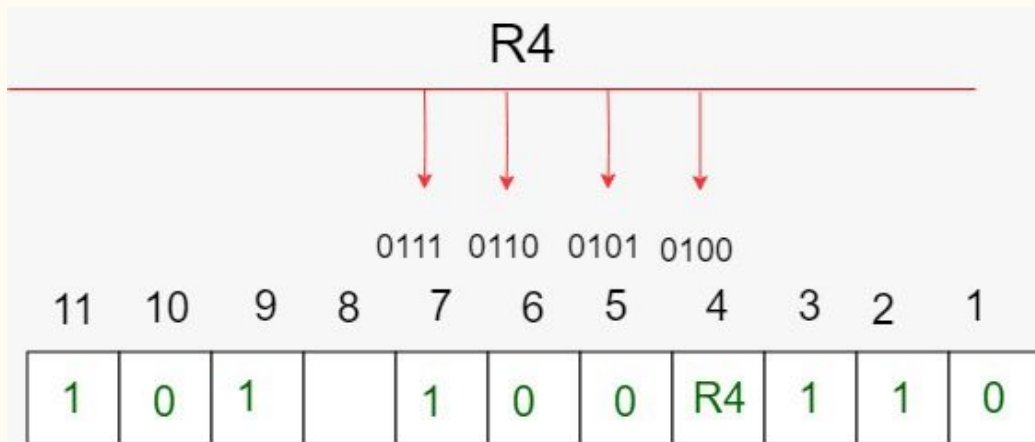
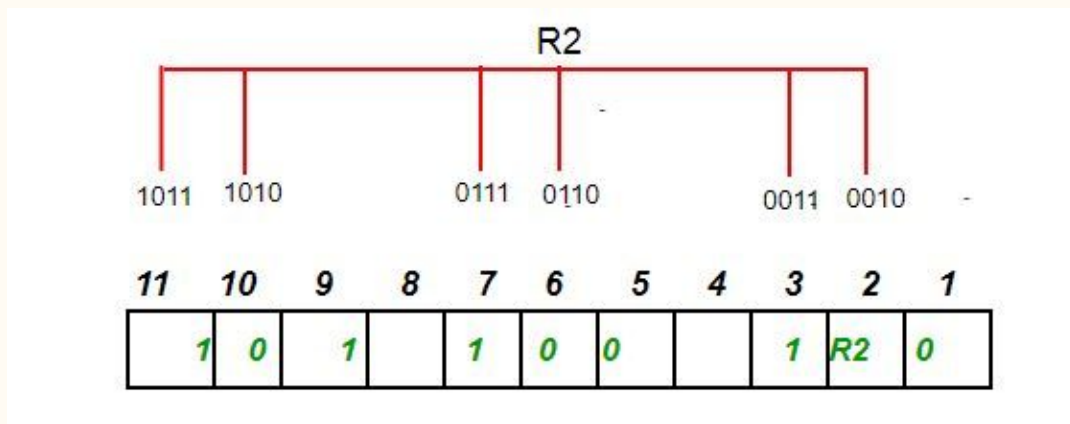
Suppose the data to be transmitted is 1011001 from sender to receiver, the bits will be placed as follows:

11	10	9	8	7	6	5	4	3	2	1
1	0	1	R8	1	0	0	R4	1	R2	R1

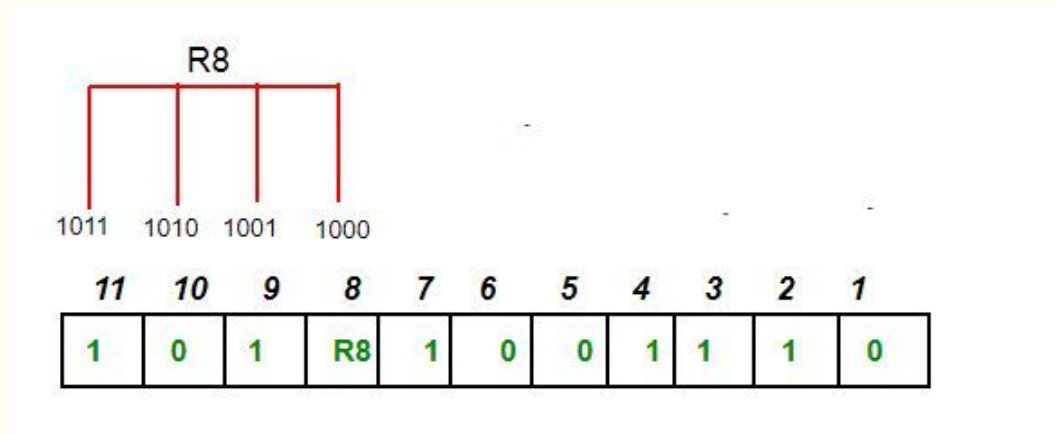
Determining The Parity Bits According to Even Parity



Error Correction: Hamming Code



Error Correction: Hamming Code



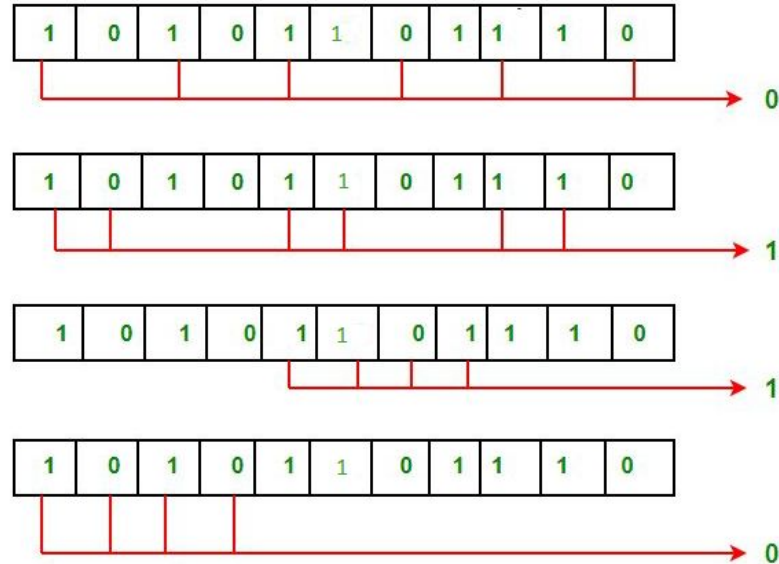
Data to be transferred:

11	10	9	8	7	6	5	4	3	2	1
1	0	1	0	1	0	0	1	1	1	0

Error Correction: Hamming Code

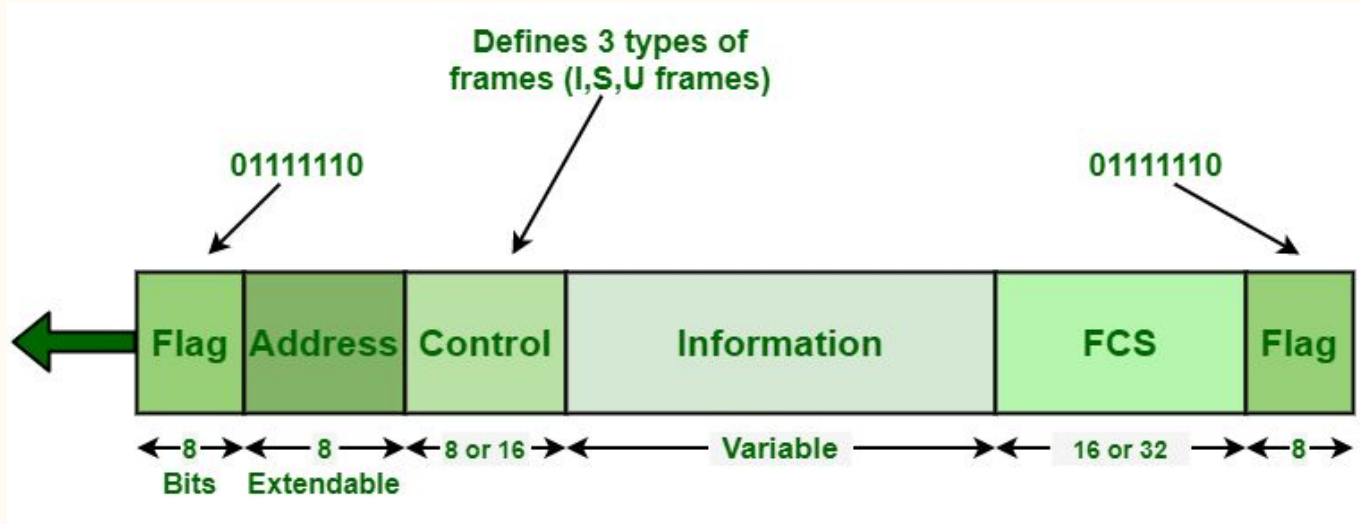
Suppose 6th bit is flipped:

The bits give the binary number 0110 whose decimal representation is 6. Thus, bit 6 contains an error. To correct the error the 6th bit is changed from 1 to 0.



Protocol: High-Level Data Link Control (HDLC)

A high-level data link control (HDLC) is a protocol that is a **bit-oriented synchronous** data link layer protocol. HDLC ensures the error-free transmission of data to the proper destinations and controls the data transmission speed. HDLCs can provide both connection-oriented and connectionless services.



High-Level Data Link Control (HDLC)

HDLC frame consists of following fields:

- 1) **Flag field:** Flag field contains synchronization pattern 01111110 that identifies both the beginning and the end of a frame.
- 2) **Address field:** It contains the address of the secondary stations. If a primary station created the frame, it contains a to address. If a secondary creates the frame, it contains a from address.
- 3) **Information field:** It contains the user's data from the network layer or management information.

High-Level Data Link Control (HDLC)

4) **FCS field:** The frame check sequence (FCS) is basically the HDLC error detection field. It can contain either 2 or 4 byte.

5) **Control field:** The control field is a 1 or 2-byte segment of the frame used for flow and error control. It contains different data for each of the different types of frames i.e I frames, S frames and U frames.

HDLC Frames Types:

1. *Information frames (I frames):* Used for data transfer and control information
2. *Supervisory frames (S frames):* Used for control information
3. *Un-numbered frames (U frames):* Used for system management

High-Level Data Link Control (HDLC)

HDLC Frame

I - frame

Flag	Address	Control	User data from upper layers	FCS	Flag
------	---------	---------	-----------------------------	-----	------

S - frame

Flag	Address	Control	FCS	Flag
------	---------	---------	-----	------

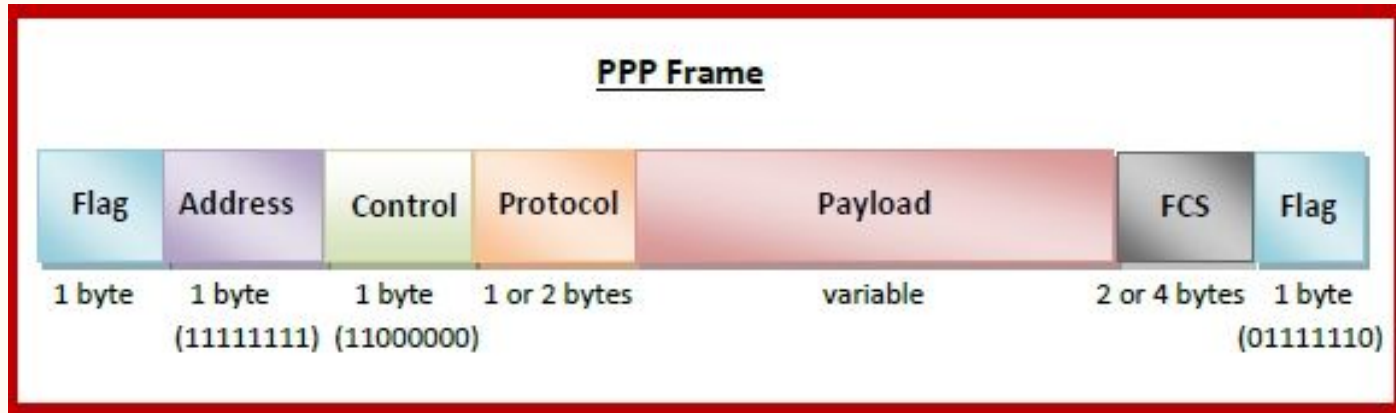
U - frame

Flag	Address	Control	Management Information	FCS	Flag
------	---------	---------	------------------------	-----	------

Protocol: Point to Point Protocol (PPP)

- The Point-to-Point Protocol (PPP) is a data link layer (Layer 2) utility designed to establish a direct connection between two nodes.
- While HDLC is often restricted to specific hardware environments, PPP is the standard for connecting home users to ISPs and interconnecting routers over a wide variety of physical media.
- Allows for the dynamic assignment of IP addresses at connection time.
- PPP includes robust mechanisms for two devices to verify each other's identity (using PAP or CHAP) before any data is exchanged, a feature entirely absent in standard HDLC.

Point to Point Protocol (PPP)



1. Flag field: Flag field marks the beginning and end of the PPP frame. Flag byte is 01111110. It is of 1 byte.

2. Address field: This field is of 1 byte and is always 11111111. This address is the broadcast address i.e. all the stations accept this frame.

Point to Point Protocol (PPP)

3. Control field: This field is also of 1 byte. This field uses the format of the U-frame (unnumbered) in HDLC. The value is always 00000011 to show that the frame does not contain any sequence numbers and there is no flow control or error control.

4. Protocol field: This field specifies the kind of packet in the data field i.e. what is being carried in data field.

5. Data field / Payload: Its length is variable. If the length is not negotiated using LCP during line set up, a default length of 1500 bytes is used. It carries user data or other information.

6. FCS field: The frame checks sequence. It is either of 2 bytes or 4 bytes. It contains the checksum.

Channel Allocation Problem

- If two transmitters transmit at the same time, their signal may interfere or collide, a method is needed to share the broadcast link among the various transmitters and avoid such collision.
- The channel allocation problem is **"How to allocate a single broadcast channel among competing users"**.
- There are a variety of solutions to the problem of allocating multi-access channels among multiple competing users. They are broadly classified as: ***Static Channel Allocation and Dynamic Channel Allocation***.

Channel Allocation Problem

- **Static Channel Allocation:** The channel's capacity is essentially divided into fixed portions; each user is then allocated a portion for all time.
Example: *FDM and TDM (Frequency/Time Division Multiplexing)*.
FDM is used in Radio or TV broadcasting whereas TDM is POTS (Plain Old Telephone System).
These channel allocation techniques waste bandwidth as they cannot cope with the dynamic environment.
- **Dynamic Channel Allocation:** With a dynamic approach the allocation of the channel [is] based on the traffic generated by the users. It tries to get better utilization of the channel when traffic is unpredictable.
Example: *Pure/Slotted ALOHA Protocol or Carrier Sense Multiple Access (CSMA) Protocols, etc.*

Random access Protocols: ALOHA

ALOHA is an early random access protocol used in computer networks. It was first used in the ALOHAnet network at the University of Hawaii in the 1970s. The basic operation of the ALOHA protocol is as follows:

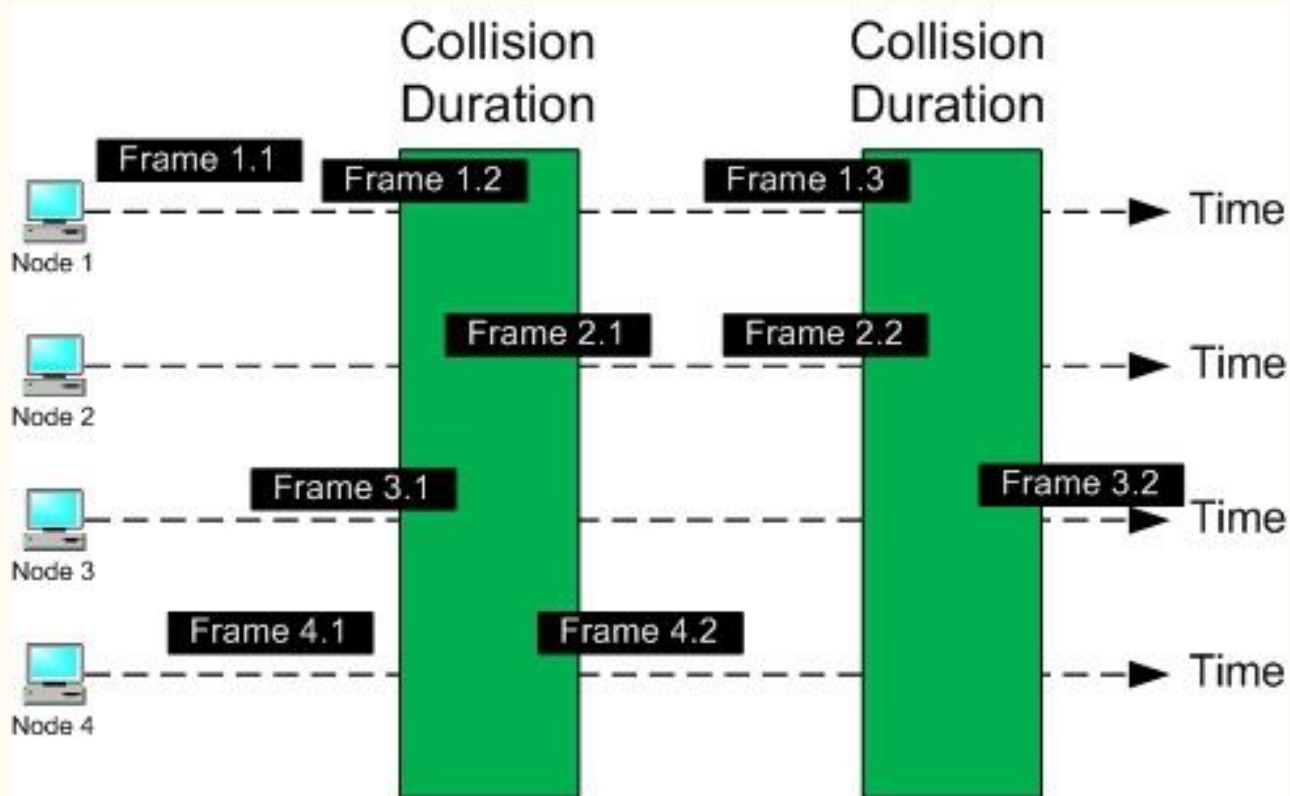
- Devices can transmit data whenever they have a message to send.
- If two or more devices transmit simultaneously, their messages will collide and be corrupted.
- Devices that detect a collision will wait for a random amount of time before trying to transmit again.

*ALOHA has two variants: **Pure ALOHA**, **Slotted Aloha**.*

Pure ALOHA

- In pure ALOHA, the stations transmit frames whenever they have data to send and expects ACK from receiver.
- When two or more stations transmit simultaneously, there is collision and the frames are destroyed.
- If acknowledgement is not received within specified time, the station assumes that the frame (or acknowledgement) has been destroyed.
- To resend the frames, it waits for a random amount of time to prevent same frames from colliding again and again.
- The probability of successful transmission is $S = G \cdot e^{-2G}$. (where G = no. of stations)
- The maximum throughput occurs at $G = 1/2$ which is 18 % .

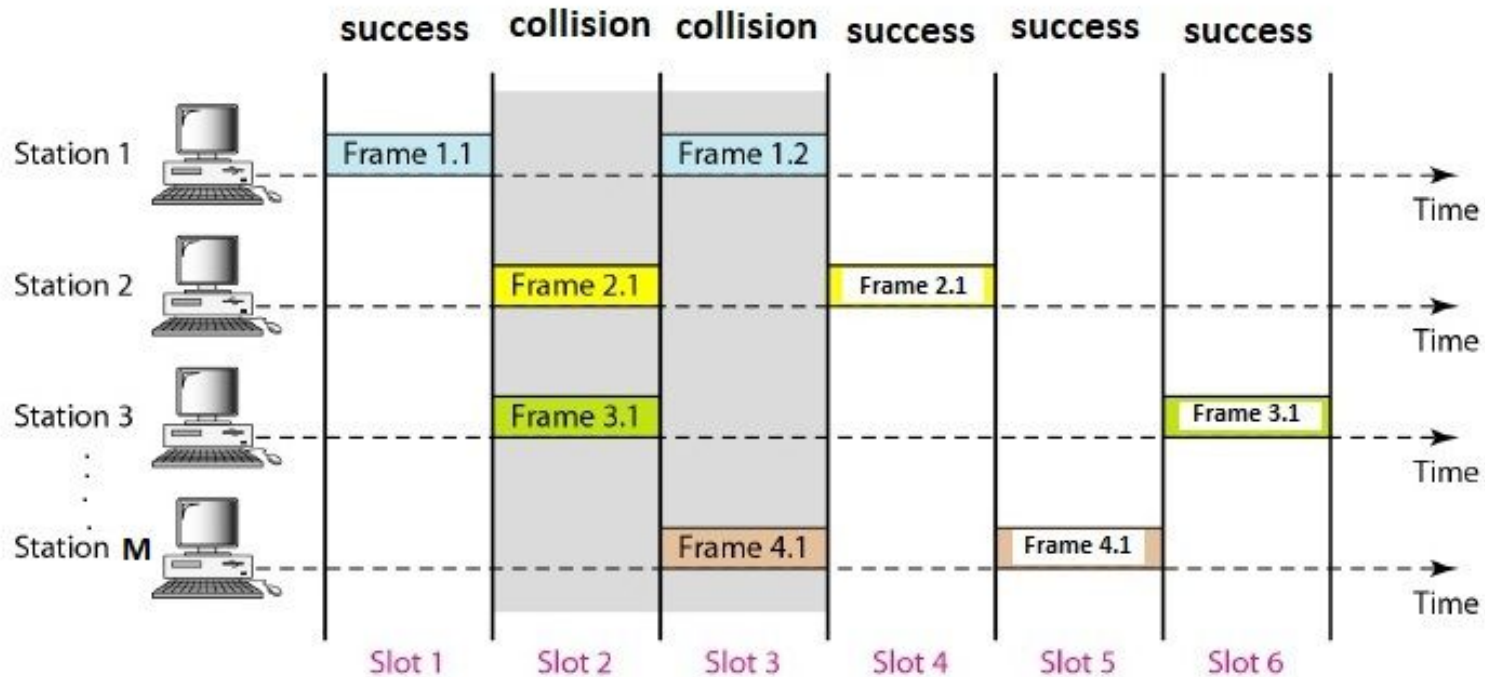
Pure ALOHA



Slotted ALOHA

- In slotted ALOHA, the time of the shared channel is divided into discrete intervals called slots and each station can send a single frame only at the beginning of the slot.
- If any station is not able to place the frame onto the channel at the beginning of the slot i.e. it misses the time slot, then the station has to wait until the beginning of the next time slot.
- There is still a possibility of collision if two stations try to send at the beginning of the same time slot.
- The probability of successful transmission is $S = G \cdot e^{-G}$ and the maximum throughput occurs at $G=1$ which is 37%.

Slotted ALOHA



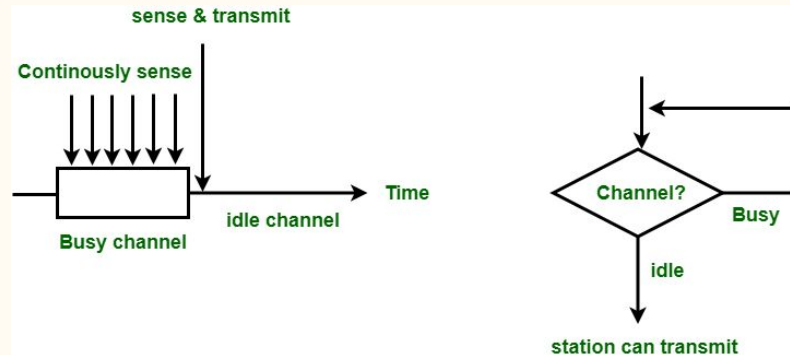
Random access Protocols: CSMA

Carrier Sense Multiple Access (CSMA) is a network protocol used to manage how devices share a communication channel to avoid data collisions.

- Instead of just sending data, a station listens to the cable first to see if anyone else is transmitting.
- If the line is idle, the station starts sending. If it's busy, the station waits its turn until the line clears up.
- Even with sensing, collisions still happen. This is because it takes a tiny amount of time for a signal to travel across the wire(Propagation delay).
- Station A might start sending because the line looks clear. But because A's signal hasn't reached Station B yet, B also thinks the line is clear and starts sending too.
- Since both stations are transmitting at the same time, their data collides and gets garbled.

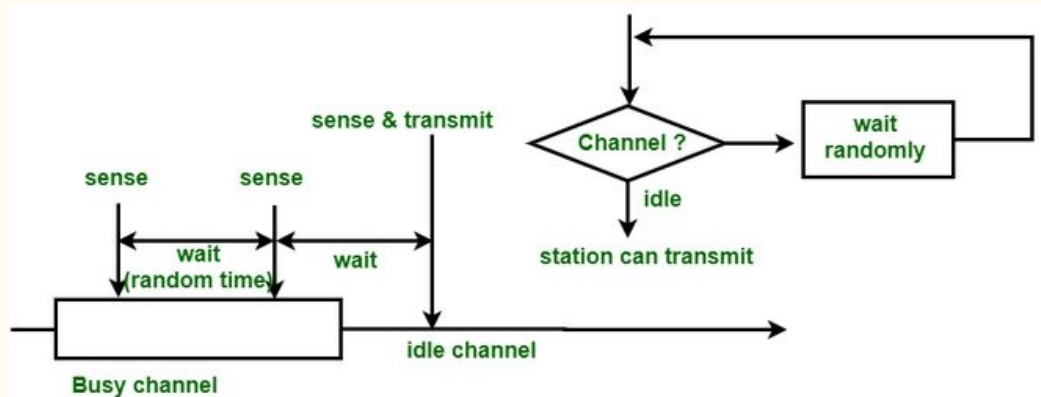
Persistent CSMA

- The device listens to the channel before transmitting.
- If the channel is free, it transmits immediately.
- If the channel is busy, the device keeps listening until the channel becomes free, then transmits right away.
- **Advantage:** Less time is wasted compared to non-persistent CSMA because there's no random delay.
- **Disadvantage:** Higher risk of collisions because multiple devices waiting for the channel to clear may all transmit simultaneously once it's free.



Non-Persistent CSMA

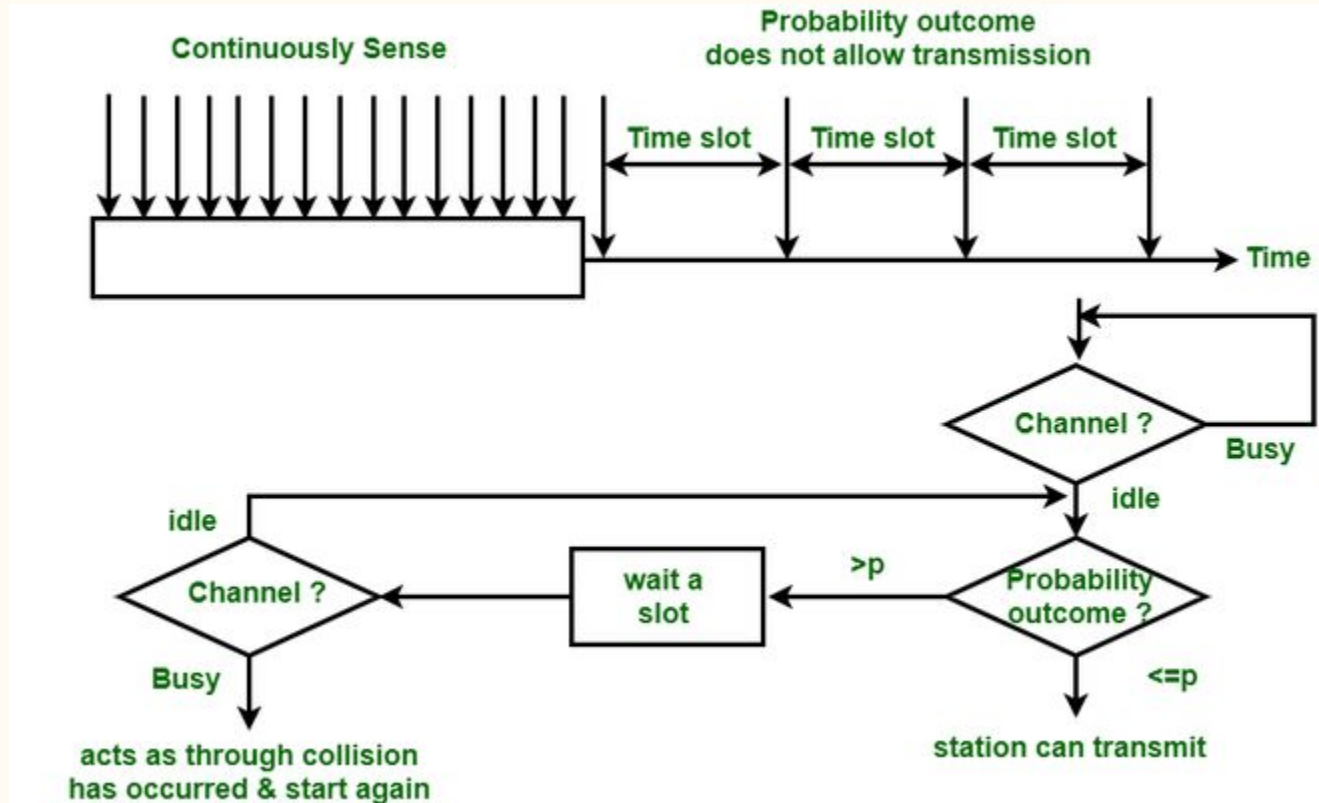
- When a device is ready to send a frame, it checks the channel.
- If the channel is free, it transmits immediately.
- If the channel is busy, the device waits for a random amount of time before checking again.
- **Advantage:** Reduces the likelihood of collisions since devices don't rush to transmit as soon as the channel is free.
- **Disadvantage:** Can lead to longer delays due to random waiting periods



P-Persistent CSMA

- When a station is ready to send data, it first checks the channel.
- If the channel is busy, the station waits for the next time slot.
- If the channel is idle :The station will transmit the frame with probability p and wait for the next time slot with probability $q = 1 - p$.
- **Advantage:** Reduces collision likelihood compared to persistent CSMA because not all devices transmit immediately when the channel is free.
- **Disadvantage:** Increases transmission delay since a device might choose not to transmit even when the channel is free.

P-Persistent CSMA



Random access protocols: CSMA/CD

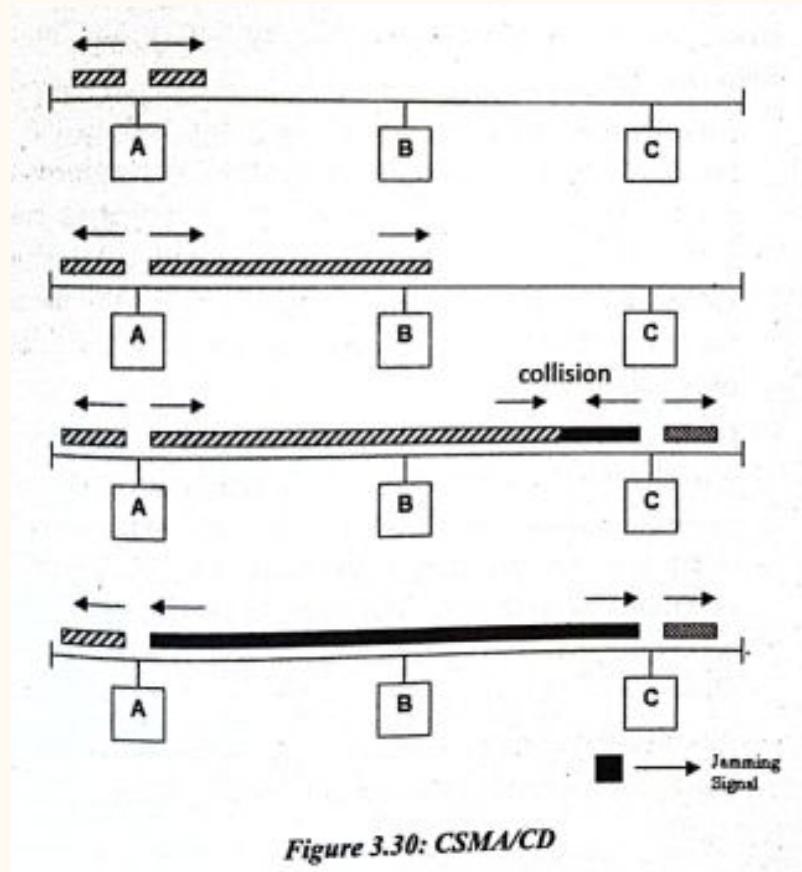
Carrier Sense Multiple Access with Collision Detection (CSMA/CD) is a network protocol for carrier transmission that operates in the Medium Access Control (MAC) layer.

- A station listens to the shared channel first. It transmits if channel is idle else waits.
- While the station is sending its data, it monitors the line to see if its own signal gets garbled by someone else's.
- If the station detects a distorted signal (a collision), it knows another station started transmitting at roughly the same time.
- The station stops sending its actual data and instead sends out a specific jam signal. This ensures every other station on the network knows a collision happened and stops too.

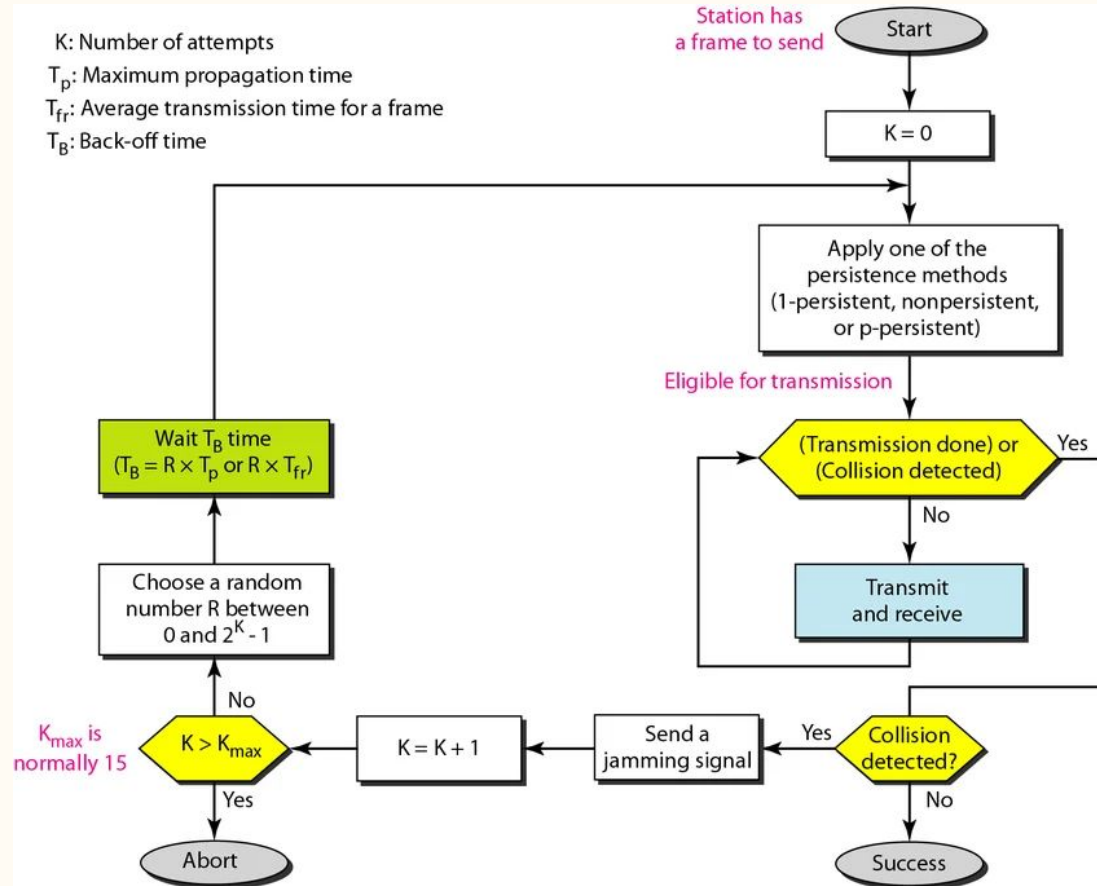
Random access protocols: CSMA/CD

- To prevent the two stations from crashing into each other again immediately, they each wait for a random amount of time before trying to listen and send again.
- **Advantages**
 - Saves Bandwidth: It kills a collided transmission immediately so you don't waste time sending garbage.
 - Efficient: Much smarter than ALOHA because it actually listens first.
- **Disadvantages**
 - Wired Only: It doesn't work for Wi-Fi because you can't talk and listen at the same time.
 - Distance Limits: If the cable is too long, the "crash" signal won't reach the sender in time.
 - Crowded: If too many people try to talk at once, the network spends all its time "backing off" instead of working.

Random access protocols: CSMA/CD



Random access protocols: CSMA/CD



Random access protocols: CSMA/CA

Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) is a medium access control (MAC) protocol that prevents data collisions in wireless communication. In contrast to CSMA/CD that deals with collisions after their occurrence, CSMA/CA prevents collisions prior to their occurrence.

- Before sending anything, the station listens to the channel. If it's busy, it waits.
- Even if the line is quiet, the station doesn't jump in immediately. It waits for a tiny period called Inter-Frame Space (IFS) to make sure the previous transmission is actually finished.

Random access protocols: CSMA/CA

- After the IFS, the station picks a random back-off time. It counts down only while the line is clear. This stops multiple waiting stations from all shouting the moment the line opens up.
- To be extra safe, the sender can send a "Request to Send" (RTS). If the receiver is ready, it hits back with a "Clear to Send" (CTS). This tells everyone else nearby to be quiet for a specific duration.
- Unlike Ethernet, Wi-Fi requires an explicit ACK for every frame. If the sender doesn't get one, it assumes a collision happened and starts the whole process over.

Random access protocols: CSMA/CA

- ***Advantages of CSMA/CA***

- CSMA/CA prevents collision.
- Due to acknowledgements, data is not lost unnecessarily.
- It avoids wasteful transmission.
- It is very much suited for wireless transmissions.

- ***Disadvantages of CSMA/CA***

- The algorithm calls for long waiting times.
- It has high power consumption

Controlled Access Methods

Controlled access is a deterministic protocol where stations consult one another or a central authority to find which station has the right to send.

Unlike random access methods, these protocols ensure that no two stations transmit simultaneously, thereby completely eliminating data collisions.

Primary Categories:

Reservation: Stations "book" a slot before sending.

Polling: A primary station coordinates all transmissions.

Token Passing: A logical ring determines transmission rights via a special frame.

Controlled Access Methods: Reservation

This method operates in two distinct phases: the *Reservation Phase* and the *Data Transmission Phase*.

Reservation Phase:

Time is divided into intervals. Each interval begins with a reservation frame consisting of N short slots, where N is the total number of stations in the network. If Station i has a frame to send, it transmits a '1' bit in the i th slot. If it has no data, it leaves the slot as '0'.

Data Phase:

After all N slots are checked, the stations that signaled a '1' transmit their full data frames in numerical order.

Controlled Access Methods: Polling

This method is designed for topologies where one device is the Primary Station (Controller) and all other devices are Secondary Stations.

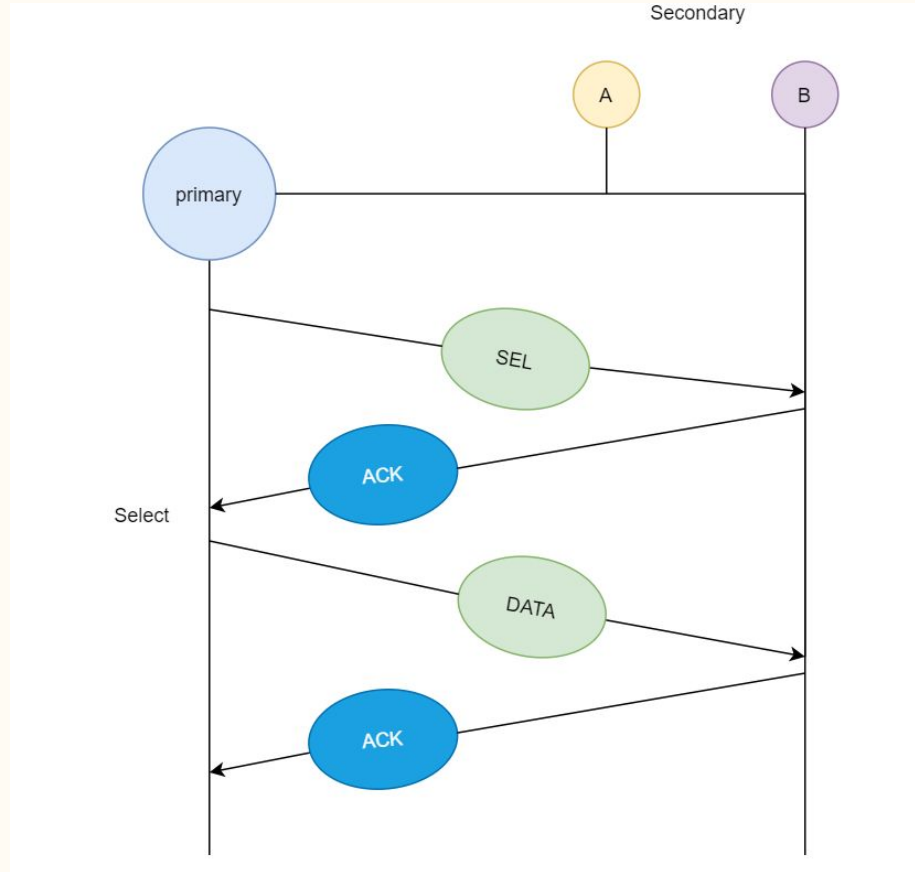
The Poll Procedure (Inbound Traffic):

- The Primary station sends a "Poll" frame to a specific Secondary.
- If the Secondary has data, it responds by transmitting its frame.
- If the Secondary is idle, it returns a NAK (Negative Acknowledgement), and the Primary moves to the next station in the polling list.

The Select Procedure (Outbound Traffic):

- When the Primary has data for a Secondary, it sends a "Select" frame to verify if the Secondary is ready to receive.
- The Secondary must acknowledge its readiness before the Primary begins the actual data transfer.

Controlled Access Methods: Polling



Controlled Access Methods: Token Passing

In this distributed method, the right to access the medium is governed by a special, small control frame called a Token.

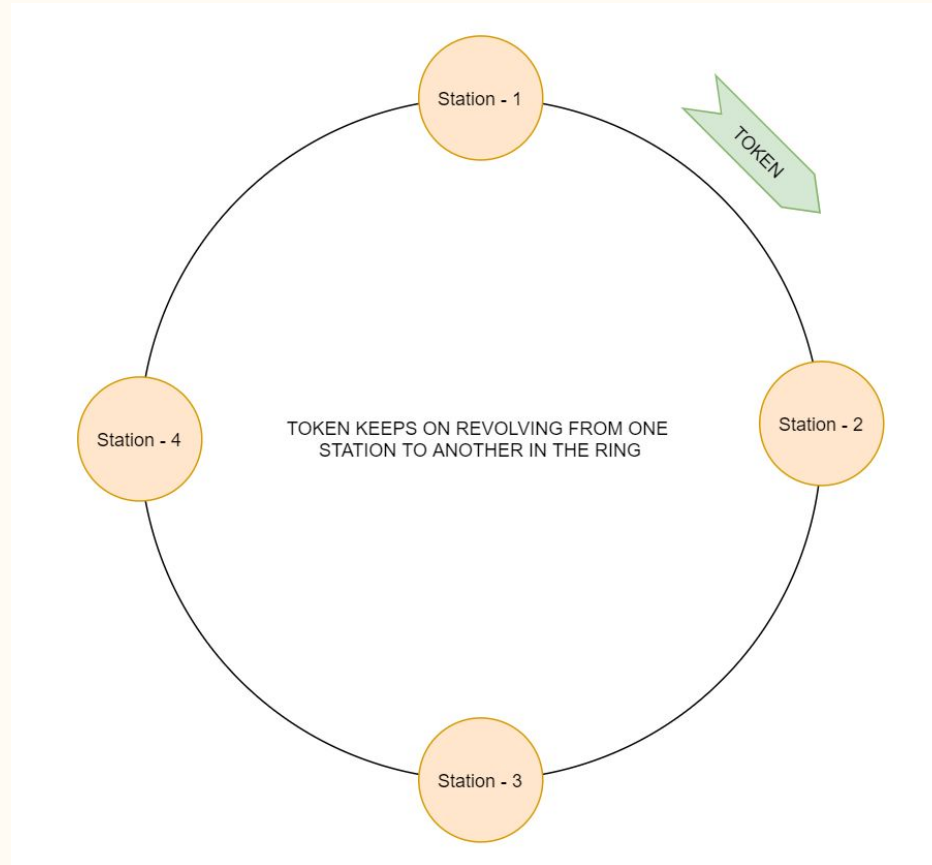
The Logical Ring: Stations are connected in a logical sequence where each station knows its predecessor and successor.

The Transmission Rule:

- A station can only transmit data when it receives and "seizes" the Token from its predecessor.
- After transmitting its data frames or after a pre-set "Token Holding Time" (THT) expires, the station regenerates the Token and passes it to its successor.

The protocol requires a designated Monitor Station to handle exceptions such as a "Lost Token" (due to noise) or an "Orphan Frame" (data that circulates indefinitely because the sender failed).

Controlled Access Methods: Token Passing



Controlled Access Methods

Advantages:

- *No Collisions*: Zero bandwidth is wasted on collisions or retransmission attempts.
- *Bounded Delay*: Every station is guaranteed an opportunity to transmit within a finite time, making it ideal for time-sensitive industrial applications.
- *Stability*: Performance does not degrade sharply under high traffic loads, unlike CSMA/CD.

Disadvantages:

- *Latency*: Even if the network is idle, a station must wait for the Token or Poll to arrive before it can begin transmission.
- *Complexity*: Requires sophisticated management logic for adding/removing nodes and recovering from station failures.
- *Overhead*: Continuous exchange of control frames (Polls and Tokens) consumes a portion of the total available channel capacity.

Channelization

Channelization is a multiple-access method in which the available bandwidth of a single link is shared in time, frequency, or through code, between different stations.

Unlike random access or controlled access, channelization creates dedicated "channels" for each station, allowing simultaneous transmission without interference.

The goal is to provide a conflict-free environment by partitioning the link resources logically or physically among multiple users.

Three Primary Techniques:

- FDMA (Frequency-Division Multiple Access)
- TDMA (Time-Division Multiple Access)
- CDMA (Code-Division Multiple Access)

Channelization: Frequency-Division Multiple Access (FDMA)

- In FDMA, the total available bandwidth is divided into several smaller frequency bands, or channels.
- Each station is assigned a specific frequency band to transmit its data continuously. These bands are separated by small "Guard Bands" to prevent signal overlap and interference.
- Because each station operates on a completely different frequency, multiple stations can send data at the exact same time without their signals colliding.
- Bandwidth is wasted if a station is idle, as its assigned frequency band cannot be used by other stations in the network.

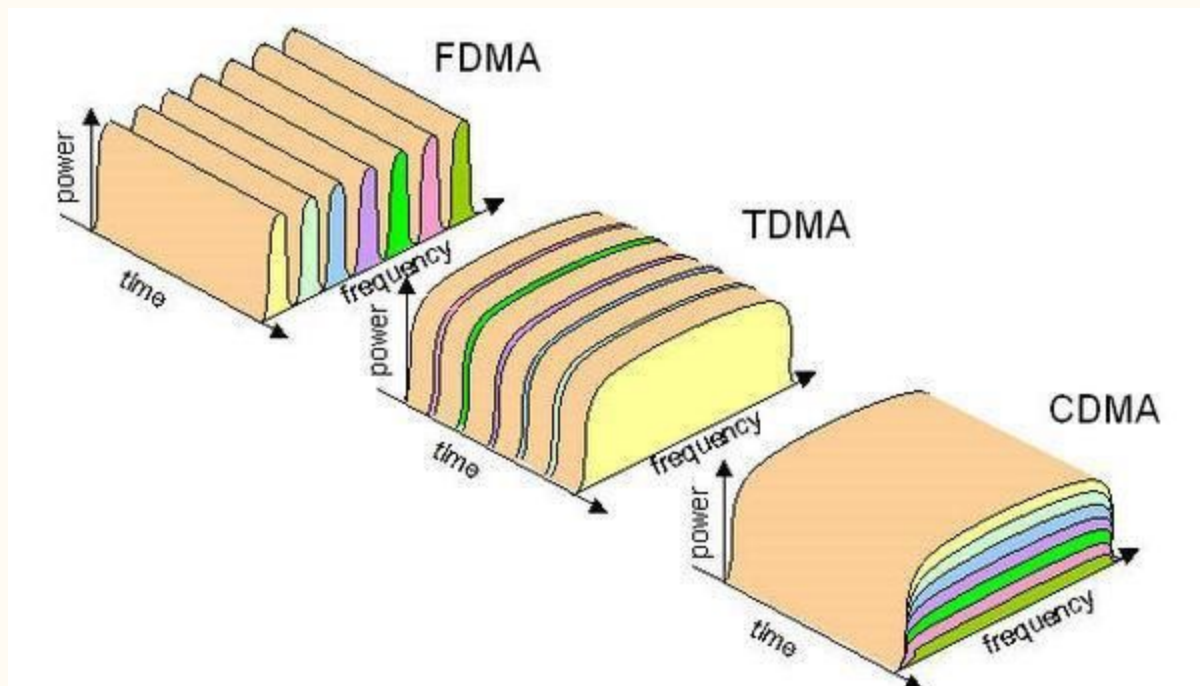
Channelization: Time-Division Multiple Access (TDMA)

- In TDMA, all stations share the same total bandwidth of the channel, but they take turns transmitting data in specific, pre-allocated time slots.
- Each station is assigned a precise time slice within a recurring "Frame." Station 1 sends during Slot 1, Station 2 during Slot 2, and so on.
- All stations must be perfectly synchronized in time to ensure that one station's transmission does not bleed into the time slot of another.
- Small gaps of silence, called Guard Times, are inserted between slots to account for synchronization errors and propagation delays.

Channelization: Code-Division Multiple Access (CDMA)

- Unlike FDMA or TDMA, CDMA allows all stations to transmit over the entire bandwidth simultaneously for the entire duration of the communication.
- Each station is assigned a unique mathematical code (a Chip Sequence or Orthogonal Code). The data is multiplied by this code before transmission.
- The receiver uses the same unique code to extract the specific data intended for it, while treating all other simultaneous transmissions as low-level background noise.
- CDMA relies on "Orthogonal Sequences" (Walsh Tables), ensuring that the mathematical product of two different codes is zero, which allows for perfect signal separation.

Channelization



Channelization

Advantages

1. *Collision-Free*: These methods provide a guaranteed path for data, making them highly reliable for constant-bit-rate applications like cellular voice calls.
2. *Predictable Performance*: Since resources are pre-allocated, the throughput and delay for each station are constant and unaffected by other users' traffic.
3. *Scalability*: CDMA, in particular, allows for high security and more efficient use of the spectrum compared to traditional analog methods.

Channelization

Disadvantages

1. *Inflexibility*: If a station has no data to send, its allocated frequency or time slot remains empty and cannot be reclaimed by others, leading to resource wastage.
2. *Technical Complexity*: TDMA requires high-precision clock synchronization, and CDMA requires complex mathematical processing and power control to function correctly.
3. *Finite Capacity*: The number of users is strictly limited by the number of available frequency bands (FDMA), time slots (TDMA), or orthogonal codes (CDMA).

Wired LAN: Ethernet

Ethernet is the most widely used wired Local Area Network (LAN) technology, defined by the IEEE 802.3 standard.

While early Ethernet used a physical bus (coaxial cable), modern Ethernet uses a physical star topology with a central switch or hub and twisted-pair cabling.

The frame format specified by IEEE 802.3 standard contains following fields:

Preamble	Destination Address	Source Address	Type	Data	Frame Check Status (FCS)
Bytes 8	6	6	2	46 to 1500 bytes	2 or 4

- **Preamble:** It is seven bytes (56 bits) that provides bit synchronization. It consists of alternating 0s and 1s. The purpose is to provide alert and timing pulse.
- **Destination Address (DA):** It is six byte field that contains physical address of packet's destination.

Wired LAN: Ethernet

The frame format specified by IEEE 802.3 standard contains following fields:

Preamble	Destination Address	Source Address	Type	Data	Frame Check Status (FCS)
Bytes 8	6	6	2	46 to 1500 bytes	2 or 4

- **Source Address (SA):** It is also a six byte field and contains the physical address of source or last device to forward the packet (most recent router to receiver).
- **Length:** This two byte field specifies the length or number of bytes in data field.
- **Data:** It can be of 46 to 1500 bytes, depending upon the type of frame and the length of the information field.
- **Frame Check Sequence (FCS):** This is for byte field, contains CRC for error detection.

Wired LAN: Ethernet

Physical Layer Specifications (10Base-X)

- *10Base5 (Thick Ethernet)*: Utilizes a thick yellow coaxial cable. It supports segments up to 500 meters and uses "vampire taps" to connect stations to the bus.
- *10Base2 (Thin Ethernet)*: Uses a thinner, more flexible coaxial cable (RG-58) with BNC connectors. It supports a maximum segment length of 185 meters.
- *10Base-T (Twisted Pair)*: The most common standard for 10 Mbps Ethernet, utilizing Category 3 or 5 UTP cables. It requires a central Hub or Switch to connect devices.

Meaning of the Name:

- * 10: Represents the data rate (10 Mbps).
- Base: Stands for "Baseband" transmission (digital signaling).
- T/2/5: Refers to the type of cable or maximum segment length.

Wired LAN: Ethernet

Fast and Gigabit Ethernet

- *Fast Ethernet (802.3u)*: Increases the data rate to 100 Mbps while maintaining the same frame format as Standard Ethernet. It primarily uses 100Base-TX (Category 5 UTP).
- *Gigabit Ethernet (802.3z / 802.3ab)*: Provides a transmission rate of 1000 Mbps. It supports both fiber optic cables and Category 5e/6 UTP cables.

Wired LAN: FDDI (Fiber Distributed Data Interface (FDDI))

- FDDI is a high-speed Local Area Network (LAN) standard developed by ANSI (X3T9.5) that utilizes fiber-optic cabling to transmit data at a rate of 100 Mbps.
- It is designed to support up to 500 stations over a total fiber distance of 200 kilometers, making it ideal for campus-wide "backbone" networks.
- Access to the shared medium is controlled through a deterministic Token Passing mechanism, which prevents data collisions.
- While primarily using multi-mode fiber optics, the standard also supports twisted-pair copper cabling (known as CDDI).

Wired LAN: FDDI (Fiber Distributed Data Interface (FDDI))

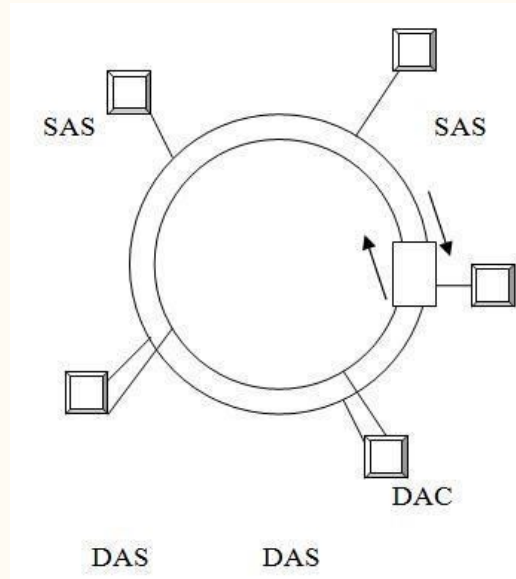
Dual Ring Architecture:

- FDDI employs a dual-ring topology consisting of two independent counter-rotating rings: the Primary Ring and the Secondary Ring.
- Under normal operating conditions, data frames travel only on the Primary Ring, while the Secondary Ring remains idle as a standby resource.
- The rings are "counter-rotating," meaning the signal travels clockwise on one ring and counter-clockwise on the other.
- If a link fails or a cable is cut, the stations on either side of the break detect the loss of signal.
- The neighboring stations automatically "wrap" the Primary Ring into the Secondary Ring, creating a single, continuous logical loop.
- Once the wrap occurs, the failed segment is isolated from the rest of the network for maintenance without causing a total system shutdown.

Wired LAN: FDDI (Fiber Distributed Data Interface (FDDI))

Dual Attachment Station (DAS): Connected to both rings for maximum reliability.

Single Attachment Station (SAS): Connected only to the Primary Ring via a concentrator.



Wired LAN: FDDI (Fiber Distributed Data Interface (FDDI))

FDDI Frame:

FDDI Frame format:

Preamble	SD	FC	DA	SA	Data	Checksum	ED	FS
8 B	1 B	1 B	2 or 6 B	2 or 6 B	4500 B	4 B	1 B	1 B

Preamble: Unique sequence that prepares each station for an upcoming frame.

Start Delimiter: Indicates beginning of the frame.

Frame Control: Indicates size of address field and whether the frame contains synchronous or asynchronous data, among other control information

SA/DA: Source and Destination

Data: information destined for upper layers or control information

Frame Check Sequence: For Error

End Delimiter: End of Frame.

Frame status: Allows the source station to determine whether an error occurred; identifies whether the frame was recognized and copied by a receiving station.

Wireless LAN (IEEE 802.11)

- A Wireless LAN (WLAN) is a flexible data communication system that uses radio frequency (RF) technology to transmit and receive data over the air, minimizing the need for wired connections.
- This is the industry-standard blueprint for WLANs, commonly referred to as Wi-Fi, defining both the Physical and Data Link layers.
- **Core Components:**
 1. *Station (STA)*: Any computing device with a wireless network interface (e.g., Laptops, Smartphones).
 2. *Access Point (AP)*: A bridge device that connects wireless stations to a wired "Distribution System" (Ethernet).

Basic Service Set (BSS): The smallest unit of a WLAN, consisting of a group of stations and an optional Access Point.

Wireless LAN (IEEE 802.11)

Architecture:

Ad-Hoc Mode (Independent BSS):

Stations communicate directly in a peer-to-peer fashion without a central Access Point.

Use Case: Quick, temporary file sharing between devices in close proximity.

Infrastructure Mode:

All stations communicate via a central Access Point. Even local traffic between two wireless nodes must pass through the AP.

Wireless LAN (IEEE 802.11)

Extended Service Set (ESS): Created by linking multiple BSSs through a wired backbone, allowing users to "roam" across a building without losing their connection.

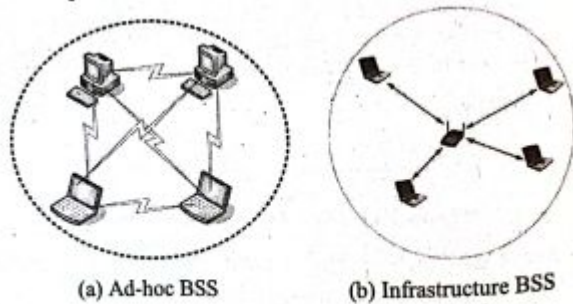


Figure 3.39: BSS

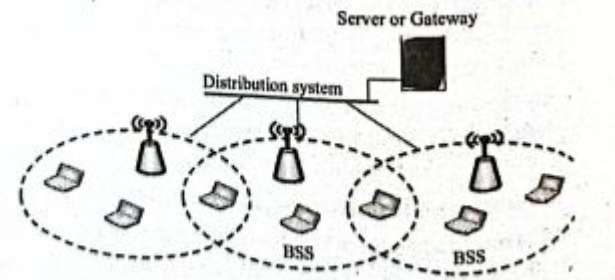


Figure 3.40: ESS

Wireless LAN (IEEE 802.11)

Standard	Frequency Band	Max Data Rate	Key Characteristic
802.11b	2.4 GHz	11 Mbps	Original popular standard; prone to interference from household electronics.
802.11g	2.4 GHz	54 Mbps	Backwards compatible with 802.11b; uses OFDM for higher speeds.
802.11n	2.4 / 5 GHz	600 Mbps	Introduced MIMO (Multiple-Input Multiple-Output) using multiple antennas.
802.11ac	5 GHz	1.3+ Gbps	Uses wider channels and beamforming to achieve fiber-like speeds wirelessly.

Wireless LAN Bluetooth Standards (IEEE 802.15.1)

- Bluetooth is a short-range wireless communication technology designed to create Personal Area Networks (PANs) by replacing cables between portable devices.
- It is officially defined by the IEEE 802.15.1 standard and operates in the unlicensed 2.4 GHz Industrial, Scientific, and Medical (ISM) band.
- The technology is optimized for low power consumption, low cost, and a small physical footprint, making it ideal for battery-operated mobile devices.
- Typically offers a communication range of 10 meters (Class 2) with data rates ranging from 1 Mbps (Basic Rate) to 3 Mbps (Enhanced Data Rate).

Wireless LAN Bluetooth Standards (IEEE 802.15.1)

Network Architecture: The Piconet

- A Bluetooth network is called a **Piconet**. It is a centralized architecture consisting of one Primary Station (Master) and up to seven active Secondary Stations (Slaves).
- The Primary device dictates the communication "beat." It defines the clock and the specific frequency-hopping sequence that all Secondary devices must follow to stay in sync.
- Up to 255 additional devices can be "parked" in a Piconet. They remain synchronized to the clock but cannot communicate until the Primary transitions them to an active state.
- A **Scatternet** is formed when multiple piconets are linked together to extend the range or capacity of the network. A single device can participate in two different piconets simultaneously.

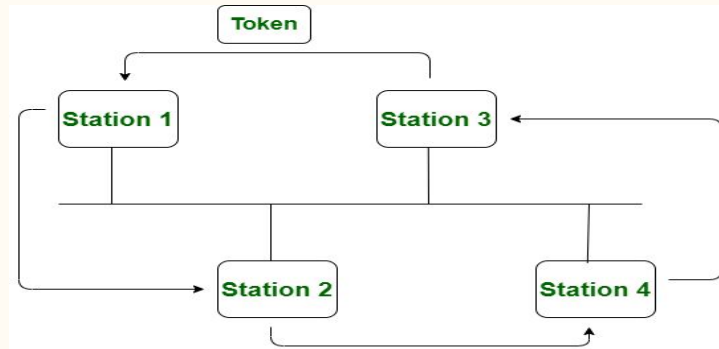
Wireless LAN Bluetooth Standards (IEEE 802.15.1)

Working:

- The 2.4 GHz band is extremely noisy because it is shared with Wi-Fi, microwaves, and other electronic devices.
- Bluetooth uses Frequency Hopping Spread Spectrum (FHSS) to maintain a stable link.
- The bandwidth is divided into 79 channels, each 1 MHz wide.
- Instead of staying on one frequency, the Primary and Secondary stations "hop" to a new frequency 1,600 times every second.
- If a specific frequency encounter interference (noise), the packet is lost. However, the very next packet is sent on a completely different frequency, ensuring the interference only affects a tiny fraction of the transmission.

Token Bus (IEEE 802.4)

- Stations are connected to a single, common coaxial cable (a Linear Bus).
- Although physically a bus, the stations are organized into a Logical Ring. Each station is assigned a position in a sequence based on its numeric address.
 - The Token is passed from the station with the highest address to the station with the next lower address.
 - The last station in the sequence passes the Token back to the first, completing the logical circle.

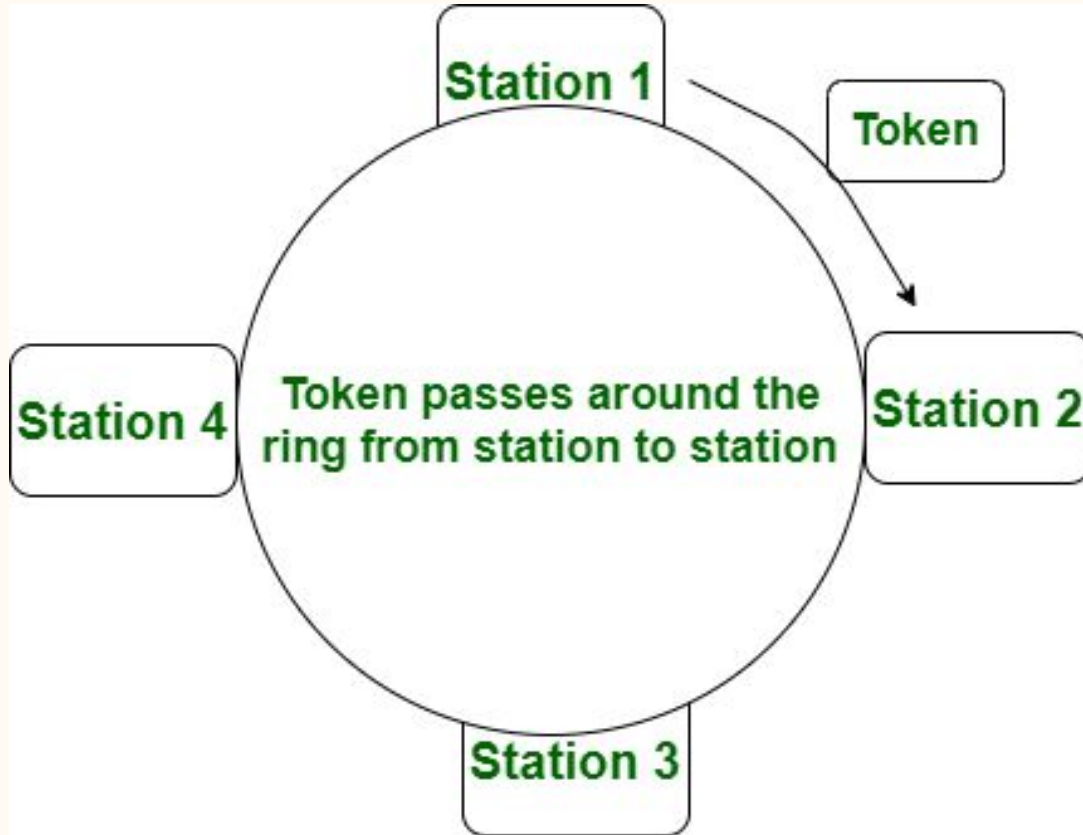


Token Ring (IEEE 802.5)

Stations are connected in a physical circle where each station is directly linked to its immediate neighbors (Predecessor and Successor).

- A station waiting to send data waits for a "Free Token" to arrive. It seizes the token and changes a single bit, converting it into a "Busy" start-of-frame.
- The station appends its data and the destination address to the token and sends it forward.
- The destination station copies the data and marks the frame as "received."
- The frame completes the full circle back to the sender. The sender removes the data and releases a "Free Token" back onto the ring.

Token Ring (IEEE 802.5)



VLAN : How a LAN Works

- Devices are physically connected using guided media, primarily Category 6 UTP cables or Fiber Optics, forming a physical star or bus topology.
- Every device is identified by a unique 48-bit MAC Address. The network uses these addresses to deliver frames to the correct hardware interface.
- In modern LANs, a Network Switch acts as the intelligent controller. It "learns" which MAC address is on which port and forwards data only to the intended recipient.
- By default, all devices in a LAN are in the same Broadcast Domain, meaning a broadcast message (like an ARP request) is sent to every device on the network.

VLAN : How a LAN Works

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VLAN

- A Virtual LAN (VLAN) is a logical grouping of network users and resources that are grouped together regardless of their physical location on the network.
- VLANs allow a network administrator to segment a single physical switch into multiple logical switches.
- Each port on a switch is assigned to a specific VLAN ID. Traffic from a port in VLAN 10 is logically isolated from ports in VLAN 20, even though they share the same physical hardware.
- When data moves between switches, a 4-byte header (Tag) is added to the Ethernet frame to identify its VLAN membership.

VLAN Trunking

- While VLANs allow us to logically separate traffic within a single switch, Trunking is the essential mechanism that allows these multiple VLANs to travel across a single physical link between different switches.
- It acts as a high-speed 'highway' that uses tagging to ensure data from different departments stays organized and isolated as it moves across the network backbone

VLAN

